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A Critical Review of Heat Pipe Experiments in Nuclear Energy Applications

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Heat pipe experimental data sets that contained pertinent information that may be beneficial to researchers were provided regarding nuclear technology. Heat pipes have shown to have a tremendously positive impact on nuclear technologies and will continue to become a more prevalent technology as more nuclear reactor concepts embrace this robust technology. Most previous reviews may only focus on a specific heat pipe design or application, and some are backdated. This review article is to extend previous efforts and integrate and provide a summary of previously reported heat pipe experimental efforts and update the review with recently reported results in the literature for heat pipes in all nuclear-related applications, including space power (thermal radiators, core cooling, and electricity production), Microreactors (Emergency core cooling, hybrid controlled rods, reactor core cooling), and other heat pipe involvement in other nuclear-related technology (spent fuel pool cooling). The two main objectives of this review article: (1) to help facilitate the development of heat pipe codes by outlining some of the existing experimental data sets to validate their codes and directing developers to these efforts (2) to provide comprehensive information regarding the vast applicability of heat pipes used in the nuclear industry, including the theory of operation and limitations to supplement researchers in the development of new ideas for potential applications in nuclear-related technologies. The review clearly shows extensive and diverse experimental data sets for heat pipes developed under diverse testing conditions depending on the available nuclear application for validation purposes. Thus, the article is oriented to provide attention to the existing efforts but not to determine the gaps in heat pipe research

Keywords: heat pipes; heat pipe operational limits, space reactors; microreactors, spent fuel pool cooling

I. INTRODUCTION

The advancement of nuclear-related technologies has been primarily accelerated due to the integration of passive cooling mechanisms. By implementing autonomous operating methods to remove thermal energy from the reactor core without the external use of blowers or pumps, several integral mechanisms of a reactor system can be significantly reduced or omitted. The integration of such technology subsequently grants reactor systems the advantage of potentially being more compact than traditional reactor systems. Additionally, passive cooling systems have made it possible for reactors to passively respond to fluctuating grid conditions, allowing for more modularity in the electrical power output than established reactor systems. Passive cooling systems have been integrated into several Gen IV reactor designs; some operate using natural circulation, taking advantage of buoyancy-driven flow, while others utilize heat pipes (HPs) technology.

Heat pipes are passive phase-change cooling devices that transmit high heat transfer rates with a slight temperature difference over considerable distances. Given the various combinations of design types, heat pipes have become an integral part of advancing a broad range of technologies in recent days. This article reviews heat pipe experimental studies related to today's nuclear technological endeavors. The historical development of heat pipes, along with its historical advancements pertaining to its integration into nuclear-related technologies, are discussed. The article includes the heat pipe classification, the general principles of operation, and heat transfer limitations in nuclear applications. Heat pipe experiments that accelerated the advancement of nuclear power in space reactors, microreactors, and other nuclear-related technologies are summarized. Experimental data sets containing pertinent information that may be beneficial to researchers will also be provided.

The two main objectives of the review article are (1) to help facilitate the development of heat pipe codes by providing researchers with experimental data sets containing sufficient data to validate their codes and (2) to provide comprehensive information regarding the vast applicability of heat pipes used in the nuclear industry to supplement researchers in the development of new ideas for potential applications in nuclear-related technologies. Several experimental data sets and their working fluid(s), operating temperature ranges, wall material, wick design, dimensions, and heat power input will be provided. The operational methods for heating and cooling for each data set will be detailed. For each case, either steady-state or transient axial temperature profiles will be provided under conditions specific to the given experiment(s). The temperature profile is the most commonly reported data in all reviewed articles. The reader could refer to the referenced article for additional data that may be available.

II. HEAT PIPE THEORY, DEVELOPMENT, and OPERATION

Given the multitude of different heat transport systems, the heat pipe has been established to be one of the most robust and efficient methods available for passively transferring large quantities of heat. However, despite the supposedly simple operational mechanics that utilize the fundamental laws of thermal hydraulics, the actual physical phenomena observed in heat pipes have been quite complex, with unique flow and heating behavior being observed for each iteration of the heat pipe. Potential complex physical phenomena observed in heat pipes stem from design and operational factors combinations. Design factors include size, structural material, wick material, wick structure (homogenous or composite), and working fluid type. Operational factors

include heating and cooling conditions, magnetism, and gravity. Combinations of both factors contribute to certain heat transfer limitations, ultimately defining its operational constraints.

Heat pipe technologies fall under two-phase closed thermosyphons, which take full advantage of the phase change occurring in the closed loop. Perkins, A. M., and his two sons first applied this advantageous heat transfer phenomenon in the mid-nineteenth century in England. Known as the Perkins tube, this device was essentially a gravity-assisted hermetically sealed tube boiler in which water is circulated in a tube between the furnace and steam drum, providing an indirect two-phase heating system (Ref. 1).

The first official documentation of capillary pressure being used as a passive driving force was conceived by R.S. Gaugler on the General Motors Corporation (Ref. 2). To provide the capillary structure, Gaugler proposed that a proper mixture of iron and ferrophosphorus powder be placed along the inner side of the tube. During the 1940s and 50s, not much attention was given to capillary-driven passive cooling mechanisms since the technology did not require sophisticated heat transfer devices.

The idea of using capillary pressure as the passive driving mechanism in a two-phase closed loop was resurrected in 1962 by Trefethen in connection with the U.S space program (Ref. 3). In 1963, a Satellite Temperature Stabilization System was proposed by Wyatt on behalf of the U.S. space program. In the patent, the cavities within the blades, tubing, and sump are lined with a fibrous material that functions as a wick to counter large fluctuations in temperatures based on the satellite's orientation in space (Ref. 4). The first time the term “heat pipe” was mentioned was in 1964 by Grover while presenting the developmental work and manufacturability of several capillary-driven liquid-metal stainless-steel heat transfer devices built for high operational temperatures (Ref. 5). In 1966, Grover filed a patent on behalf of the U.S. Atomic Energy Commission for the first officially named heat pipe, defining the heat pipe as a device that accomplishes its function by a wick of suitable capillary structure (Ref. 6). Grover’s experimental proof of liquid-metal heat pipes garnered wide recognition in several industries, with many immediately beginning implementing such technology into compact cooling systems.

II.A. Historical Applications of Heat Pipes for Nuclear-Related Endeavours

II.A.1. Space Reactors:

The advent of heat pipe technology first sparked accelerated efforts into space reactor systems with a thermionic converter, which converts heat to electricity under gravity-free conditions. The reactor design thermally coupled several heat pipes to the nuclear core components (Ref. 7). Deverall published an orbital heat pipe experiment using Grover’s heat pipe concept, demonstrating that heat pipes can function appropriately in a zero-gravity field (Ref. 8). Using Grover’s space reactor concept, Fiebelmann developed on behalf of the European Atomic Energy Community a heat pipe cooled space reactor system in 1968 (Ref. 9). Heath, and Lantz, on behalf of NASA, presented a nuclear thermionic space power system concept that employed heat pipes as the primary cooling mechanism (Ref. 10).

In 1968, Hampel and Koopman presented work implementing passive reactivity control systems for space reactors and nuclear thermal rockets associated with the Space Nuclear Propulsion Office (Ref. 11). Niederauer presented in 1971 a split-core heat-pipe reactor concept for out-of-pile thermionic power systems, in which short Li-W reactor heat pipes penetrated the

axial reflectors to form a heat exchanger with long heat pipes which wind through the shield to thermionic diodes (Refs. 12-13). Koenig presented three heat pipe reactor concepts (Mini-Reactor, Mid-Power Range (MPR) Reactor, and Out-of-Core Thermionic Reactor) for deep space mission electric propulsion systems and electricity conversion, using molybdenum heat pipes with either sodium or lithium working fluid (Ref. 14).

Several space reactor concepts with heat pipe cooling systems were being developed and proposed by individual agencies, DOE's Office of Nuclear Energy, and NASA's Office of Aeronautics and Space Technology for the SP-100 reactor (Ref. 15). The SP-100 reactor concept was cooled by a high temperature Mo-13% Re heat pipe with lithium working fluid. Subsequent design updates were performed throughout the next decade, upgrading the design to achieve higher power (Ref. 16), employing liquid metal heat pipes with fins to reduce the mass of the heat rejection subsystem (Refs. 17-19), coupling heat pipes to Brayton or Stirling power conversion subsystems (Ref. 20), and optimizing system design (Ref. 21-22)

Between the 1990s and early 2010s, several space reactor concepts utilizing heat pipe technology were proposed. Powell developed a small excore heat-pipe-thermionic space nuclear reactor power system (SEHPTR) to power spacecraft (Ref. 23). Ranken and Turner developed the Moderated Heat Pipe Thermionic Reactor (MOHTR) (Refs. 24-25). Begg presented a 40 kWe Space Thermionic Advanced Reactor-Compact (STAR-C) response to evolving Air Force space power requirements, implementing heat pipes into the radiator system (Ref. 26). Determan and Hagelston developed the Heat Pipe Cooled Thermionic reactor (HPTI), which relies on in-core sodium heat pipes to provide redundant cooling (Ref. 27). Ranken and Houts presented three reactor concepts (HPR-I, HPR-II, HPR-II) in which heat pipes couple nuclear heat to direct energy conversion systems with individual heat rejection fins directly mounted on Na/Nb heat pipes. (Ref. 28). Houts unveiled the Heat pipe Power System (HPS) concept, which provides a fission power system suitable for near-term, low-cost lunar, and planetary bases (Ref. 29). Poston and Houts detailed the HPS alongside the Heat pipe Bimodal System (HBS), which can provide substantial power and/or propulsion at low mass with high safety and reliability (Ref. 30). Poston presented the Heat Pipe-Oriented Mars Exploration Reactor (HOMER), which was a derivative of the HPS designed explicitly for producing electricity on the surface of Mars (Ref. 31). Over the mid-1990s to mid-2000s, NASA presented several Safe Affordable Fission Engine (SAFE) reactor concepts for electricity production in space, in which all prototypes utilized modular alkali metal heat pipes (Refs. 32-35). The most notable of these reactor concepts were the SAFE-400 concept, a 400-kWt reactor designed to couple with a 100-kWe Brayton power system using Na/Mo heat pipes (Ref. 33). El-Genk and Tournier presented the SAIRS (Scalable AMTEC Integrated Reactor Space Power System) concept, which is a fast-spectrum reactor cooled by sodium heat pipes, C-C armored potassium heat pipe radiators, and alkali metal thermal-to-electric conversion (AMTEC) units (Ref. 36). Schmitz presented Inconel heat pipes interfaced with the Stirling converter and lithium reactor coolant for a nuclear-Stirling power plant for the Jupiter Icy Moons Orbiter (JIMO) (Ref. 37). Tournier and El-Genk presented four radiator designs with rubidium heat pipes, each having C-C armor and fins (Ref. 38) within the launch bay of the DELTA IV Heavy vehicle and a high-temperature water heat pipe radiator with a sectored gas-cooled reactor and three Closed Brayton Cycle engines built for the DELTA-IV Heavy launch vehicle (Ref. 39). King and El-Genk presented a design combination of the Heat Pipe-Segmented Thermoelectric MultiCouple (HP-STMC) and SAIRS, known as the Submersion-Subcritical Safe Space (S4) reactor (Ref. 40-41). Hartenstine presented a radiator concept with Ti/H2O loop heat pipes for spacecraft nuclear power systems for Lunar and Martian missions (Ref. 42). Greenspan presented a preliminary assessment

of the feasibility of designing an Encapsulated Nuclear Heat Source (ENHS) reactor with a solid core from which liquid metal heat pipes remove heat (Ref. 43). Bess presented a Basic LEGO (Lunar Evolutionary Growth-Optimized) reactor design to provide Lunar surface power, which used liquid-sodium heat pipes (Ref. 44). Eades and Bettencourt developed the Space Molten Salt Reactor (SMSR) with a pill-shaped core around the heat pipes (Refs. 45-46). Yuan studied the start-up of heat pipe cooled and AMTEC conversion space reactor systems (Ref. 47-48).

In 2015, a small team of NASA and DOE experts provided a solution to enabling fission power system technology, starting with the lowest power and most cost-effective reactor series named "Kilopower." The first desired reactor design goal was a full-scale nuclear ground test, nicknamed "KRUSTY" (Kilopower Reactor Using Stirling Technology), using sodium heat pipes as the thermal transport mechanism. Titanium-water heat pipes were also used to transfer waste heat from the Stirling engines to radiator fins, which can be ejected into space (Ref. 49). Beard later updated the Kilopower reactor series by using a Hybrid screen groove and self-venting arterial heat pipes (Ref. 50-51). The Demonstration Using Flattop Fissions (DUFF) experiment was mentioned, which used an existing reactor, heat pipes, a rudimentary heat exchanger, and existing Stirling converters to produce electricity. The KRUSTY concept was also mentioned (Ref. 52). Poston most recently presented the KRUSTY nuclear system test results, which operated the power system at various temperatures and power levels for 28 consecutive hours. Testing showed that the reactor system is highly tolerant of possible failure conditions and transients (Ref. 53).

II.A.2. Microreactors:

The usage of heat pipe technology in terrestrial reactor systems did not begin until around the mid-1980s. Huebotter presented the first implementation of heat pipes into terrestrial reactors for a fast, pool-type fission reactor that utilizes a heat pipe Sodium-water heat exchanger (Ref. 54). Hampel developed an underground nuclear power station concept using a self-regulating heat pipe emplaced in the hole between the heat source and heat exchange near the surface of the Earth (Ref. 55). Ohashi presented preliminary studies on applying heat pipes in passive decay heat removal systems for the modular high temperature gas-cooled reactors (HTRs (Ref. 56). Kim presented an update for the SMART (System integrated Modular Advanced Reactor) on behalf of the Korea Atomic Energy Research Institute using heat pipes in the containment cooling system (Ref. 57). Mochizuki suggested using a heat pipe cooling system for a boiling water reactor's decay heat removal system (Ref. 58). Wang investigated a sodium heat pipe (Ref. 59) and a sodium-potassium alloy heat pipe (Ref. 60) for molten salt reactors' passive residual heat removal systems. Jeong presented a passive in-core cooling (PINC) system, proposing a hybrid heat pipe that combines heat pipes and control rods, allowing for direct decay heat removal and usage of the exact drive mechanism as control rods (Ref. 61).

Kim experimentally visualized the PINCs to observe the heat transfer performance according to flow path, fill ratios, working fluid, and mesh wick (Refs. 62-63). Kim also investigated the heat transfer characteristics and operational limitations of the pressurized hybrid heat pipe for small modular reactors (Ref. 62). Mcclure presented in association with LANL the alkali metal heat pipe fast-spectrum reactor for 1-10 MWe mobile reactor system to provide robust, self-contained, and long-term power in any environment (Ref. 64). Sviridenko presented a heat exchanger concept based on low-temperature heat pipes for the autonomous emergency Water-moderated energy reactor (WWER) (Ref. 65). Sterbentz proposed to use an array of in-core potassium heat pipes for Los Alamos National Laboratory (LANL) Special Purpose Reactor, also known as the "MegaPower" reactor (Ref. 66-67). Levinsky presented Westinghouse eVinci

Reactor for Off-Grid Markets. The eVinci reactor is an epithermal spectrum microreactor that uses sodium or potassium heat pipes as the primary heat transfer mechanism (Ref. 68). The Westinghouse eVinci heat pipe cooled reactor was also mentioned by Kennedy et al. (Ref. 69). Heat pipe-cooled microreactor concepts were analyzed computationally using MOOSE-based Multiphysics modeling and simulation software developed by Idaho National Laboratory by Sabharwall et al. (Ref. 70). Hernandez presented a micro potassium liquid-metal cooled heat pipes nuclear reactor concept similar to the Westinghouse eVinci heat pipe reactor (Ref. 71). Xie presented experimental data of a heat pipe heat exchanger applied in residual heat removal systems (Ref. 72). Cai presented a heat pipe reactor design that replaces cylindrical and hexagonal fuel elements with an annular fuel element. The heat pipe design with annular fuel was proposed to enhance the LANL-MW heat pipe reactor design (Ref. 73). Cui presented a conceptual nuclear design of a 50 kWt heat pipe cooled micro molten salt reactor (micro-MSR) (Ref. 74). Hu presented a heat pipe nuclear reactor concept employing lead-bismuth eutectic as the primary coolant. Experiments using in-core heat pipes were successfully utilized in the Passive Residual Heat Removal system during station-black-out accident scenarios without any heat pipe damage (Ref. 75). Wang presented a preliminary design of a Heat Pipe-Cooled Blanket for the CFETR (China Fusion Engineering Test Reactor), a new tokamak reactor to advance research of next-generation superconducting fusion reactor systems based on the International Thermonuclear Experimental Reactor (ITER) technologies. Gravity heat pipes instead of coolant flow channels were adopted to remove heat (Ref. 76).

II.A.3. Other nuclear-related technologies:

Ventures into implementing heat pipes into nuclear-related technology aside from direct usage in space or microreactors began around the late-2000s. Such ventures include research into nuclear reactions, nuclear seawater desalination, and spent fuel pool (SFP) waste heat recovery systems. El-Awadi presented a study to improve the performance of a cryogenic heat pipe-target system for the COSY-TOF experiment, a Time of Flight (TOF) spectrometer used in the Cooler Synchrotron (COSY) accelerator facility (Ref. 77). Very lightweight, thin liquid heat pipe-target systems containing hydrogen/deuterium were used. Nygren did a post-test examination of a Li-Ta heat pipe exposed to H plasma in the DIFFER linear plasma source Magnum PSI (Ref. 78). Challenges of using liquid-metal heat pipes in this technology were mentioned, stating that the liquid metal magneto-hydrodynamics acts like a drag force on the return flow of liquid metal from the condenser to the evaporator. Potential applications of Li-Ta heat pipes are in fusion energy experiments, specifically for linear plasma generator cooling systems (Ref. 78).

Nuclear seawater desalination is an efficient, energy-intensive method of producing potable water from seawater by using reactor heat. The steam, which can be bled off at suitable points in the secondary circuit of the nuclear power plant, can be used for the desalination process (Ref. 79). Jouhara presented opportunities for heat pipe technology in nuclear seawater desalination. By coupling heat pipes to low-temperature multi-effect distillation processes, most of the waste heat generated in various nuclear power reactors can be harnessed. The efficacy of harnessing waste heat enhances the production of larger quantities of potable water, but the integration of heat pipes can also reduce the environmental impact of the nuclear desalination process. Additionally, heat pipe-based heat recovery systems may improve the overall thermodynamics of the desalination process, ensuring that the water is contamination-free under normal processes. Adding an extra loop will also help prevent direct contact between the radiation and the produced water, which will help prevent operational failure occurrences (Ref. 80).

Several passive waste heat removal/recovery systems that use heat pipes have been proposed for SFP systems. Xiong presented a study on the thermal performance of a loop-type heat pipe for passively removing residual heat from SFPs using ammonia heat pipes. Results showed that the loop heat pipe design was feasible in passive residual heat removal systems (Refs. 81-82). Xiong also presented the results of an experimental study on the thermal energy removal capacity of the heat pipe loop for passive cooling of SFPs during accident scenarios. Results indicated that significant heat could be removed using heat pipes under conditions that may occur in SFPs (Ref. 83). Jeong investigated hybrid heat pipe-based passive cooling devices for spent nuclear fuel dry storage casks (Ref. 84). Rosidi investigated the effects of graphene-fluid on the heat pipe thermal performance for passive heat removal in nuclear SFPs. The usage of graphene nano-fluid as the working fluid showed that it could significantly enhance the heat pipe thermal performance and be used as an alternative working fluid in heat pipes for a passive cooling system in SFPs (Ref. 85). Kusuma proposed vertical, straight, wickless-heat pipes as a passive residual heat removal system for nuclear SFPs. Experimental results showed that the proposed heat pipe arrangement had good thermal performance and could be used as an alternative for a passive cooling system in SFPs (Ref. 86).

Lai proposed an entirely passive cooling for spent fuel using heat pipes. A separate heat pipe-assisted passive cooling system with a significant cooling capability of 16 MW was discussed, which focused on passive cooling in the condenser section employing natural draft air cooling (Ref. 87). Kuang presented a heat transfer performance analysis of a large-scale separate heat pipe with a built-in-tube (BTSHP) for SFP heat removal. Large-scale separate heat pipes (SHP) did not perform well in long heat transport applications. Modifying the SHP into a BTSHP by inserting a straight, smooth tube in the heat pipe evaporator section allowed for about an extra 8-28% heat transfer capacity than SHPs (Ref. 88). Choi presented the Fork-End heat pipe concept for passive air cooling of SFPs. The design consisted of a condenser at the top of the heat pipe consisting of several finned tubes for passive air cooling, while the evaporator at the bottom is a single rod submerged in the spent fuel pool (Ref. 89).

II.B. Classification of heat pipes in nuclear applications

The number of heat pipe concepts proposed since its inception is quite large due to the wide range of applicable technologies. In nuclear-related technologies, various aspects of the reactor system have observed several combinations of design factors, such as design types, structural materials, wick structures, and working fluids. Capillary-driven (conventional) liquid-metal high-temperature heat pipes containing porous wick structures are most commonly observed in nuclear-related technologies.

The anatomy of a conventional (capillary-driven) heat pipe consists of a sealed conduit with end caps, a wick structure, and a small quantity of working fluid (Fig. 1). The axial length of heat pipes consists of at least two sections, namely the evaporator and condenser sections, where the working fluid is evaporated and condensed, respectively. The majority of heat pipes also contain an adiabatic section, which separates the evaporator and condenser sections by a sufficient distance to satisfy the heat transfer limitations or necessary design constraints. The wick structure inside heat pipes transports the condensate to the evaporator section. The inside of a conventional heat pipe is filled with enough working fluid to saturate the wick with liquid.

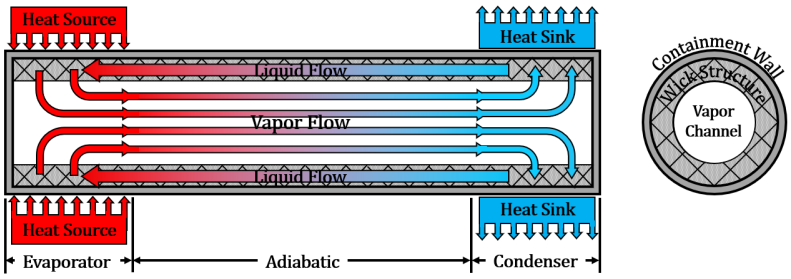


Figure 1. Radial and axial cross-section of a conventional (capillary-driven) heat pipe with incorporated heat source and heat sink.

II.B.1. Heat pipe and Wick structure material

The structural material used for a heat pipe wall is primarily determined by its application. The core variables that contribute to determining potential structural materials used for a given heat pipe are the intended operating temperature and pressure, working fluid type, environmental conditions, mode of operation, and expected operational lifetime of the application. Heat pipes in nuclear reactors are expected to handle very high temperatures and pressures and operate correctly with neutron irradiation. Additionally, the operational lifespan of a given heat pipe must exceed that of the reactor power system itself, which is typically five years (Ref. 90).

The wick structure inside heat pipes aims to provide a conduit by which the condensate can return to the evaporator section. The tiny pores of the wick are required at the liquid-vapor interface to develop enough capillary pressure to drive the working fluid. Having too small pores in the wick structure restricts the liquid from moving, leading to dryout and heat pipe failure (Ref. 91). In nuclear-related applications, the potential of dryout and heat pipe failure is a significant point of concern. Specific capillary wick structures have been implemented specifically for a given nuclear-related application to optimize the performance of the reactor system.

The screen wick (Fig. 2a) is the most commonly seen wick type in nuclear-related technologies. Most screen wick structures observed in nuclear applications are constructed by inserting single or multiple metal fabric sheets wrapped around a mandrel into the heat pipe. The mandrel is then removed, holding the wick inside the wrapped screen metal fabric (Ref. 91). The majority of screen wicks are considered homogenous, in which the wick is composed of either single or multiple layers of the same mesh number wire screen. Several homogenous screen wick types have been used in nuclear-related technologies, such as space reactors (Refs. 5, 8, 10, 16, 22, 25, 44, 93, 92-96), space heat rejection radiators (Refs. 39 and 97), hybrid heat pipe-control rods (Refs. 61, 98, and 99), reactor electronic's cooling (Refs. 100 and 101), and microreactor passive cooling systems (Refs. 74, 102-109). Composite screen wick structures, which use different mesh grades for various layers in order to capitalize on having small pores for generating high capillary pumping pressures and also large pores increasing the permeability of the liquid return, have also been utilized in applications such as space reactors (Refs. 32, 95-96) and microreactors (Ref. 108).

Sintered wicks (Fig. 2b) were the original wick structure proposed for heat pipes devices (Ref. 2). However, the thermal conductivity is higher than the screen wick since there is direct thermal contact between particles. Sintered wick structures have become popular with loop heat pipes oriented towards reactor technologies (Refs. 78, 101, 110). The sintered wick has smaller pores than screen mesh wicks, leading to a more significant capillary force that prevents the vapor

from entering the compensation chamber, decreasing the startup time and thermal resistance compared with screen wicks for the heat pipe heat exchangers and reactor cooling (Refs. 111, 112).

Axial groove wicks (Fig. 2c) are pipes with rectangular, circular, triangular, or trapezoidal grooves on the inner pipe wall. Trapezoidal grooves are the most common type. Compared with sintered and screen wicks, the size of the grooves is much larger than the porous structure, causing the capillary pumping pressure to be low and the permeability to be high relative to other wick types. Since the axial grooves are formed through either extrusion or broaching of the pipe, the effective thermal conductivity is considered relatively high. Axial groove pipes have been seen in loop heat pipe radiators for space reactor power systems (Refs. 42, 112) and space reactor cooling systems (Ref. 50).

Annulus wicks (Fig. 2d) are similar to screen wicks, except that the single-layer screen wick is placed apart from the inner wall of the heat pipe. The gap between the inner pipe wall and screen wick provides an unimpeded return flow path for the condensed liquid working fluid, making the permeability incredibly high compared with other wick types. However, the removal of direct contact between the wick and the inner pipe wall causes the effective thermal conductivity of the pipe to decrease severely. Open annulus wick types have been observed in nuclear reactor technologies, such as space reactors (Refs. 31, 33, 36, 38, 113) and microreactors (Refs. 67, 114).

Artery wicks (Fig. 2e) are screen or sintered wicks with a hollow passage that runs the pipe's length. The condensate collects inside this passage at the condenser end and is pumped back to the evaporator section by capillary forces generated at the liquid-vapor interface. Artery wicks have been seen in space reactor radiator concepts (Refs. 17 and 115) and space reactor cooling systems (Refs. 37 and 50).

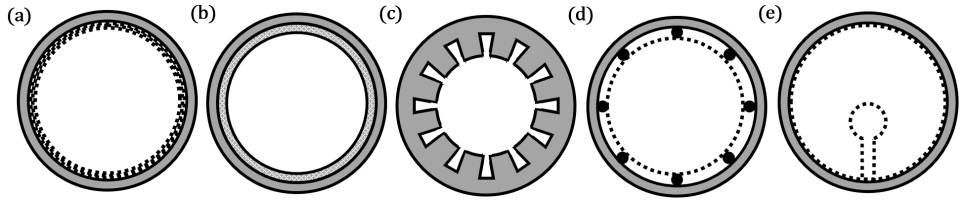


Figure 2. Basic wick structures for nuclear technologies. (a) Screen mesh wick, (b) Sintered wick, (c) Axial groove wick, (d) Open annulus wick, (e) Open artery wick.

Thermosyphons are wickless heat pipes that utilize gravity as the driving force of circulation. The condenser section is always located above the evaporator section in order for gravity to return the condensate to the evaporator section. Thermosyphons have been studied for usage in the reactor heat exchanger systems (Refs. 50, 116-120), nuclear spent fuel pool cooling systems (Refs. 85-87, 121-122), and hybrid heat pipe-control rods (Refs. 62-63, 123-124).

II.B.2. Working fluid types

Heat pipes can ultimately be categorized into four categories: cryogenic, low-temperature, moderate-temperature (or intermediate-temperature), and high-temperature heat pipes (Ref. 91). The difference between such types ultimately lies in their proper temperature ranges, maximum heat transport capabilities, and expected temperature drop at the same power level of similar geometry. The maximum heat transport capability of cryogenic (e.g., nitrogen) heat pipes is about an order of magnitude less than typical low/intermediate-temperature (e.g., ammonia) heat pipes. High-temperature heat pipes (e.g., sodium) have a maximum heat transport capacity of about three

orders of magnitude higher than cryogenic and low/intermediate-temperature heat pipes (Ref. 124-125). Given the high temperatures and substantially high heating conditions of nuclear power, high-temperature heat pipes are the most commonly observed in the direct cooling of reactor technologies. However, low/intermediate-temperature heat pipes such as two-phase closed thermosyphons have been proven helpful in removing significant amounts of decay heat from the reactor system and supplementing heat dissipation systems in space reactor radiator systems. The temperature ranges of different working fluids are provided in Fig. 3.

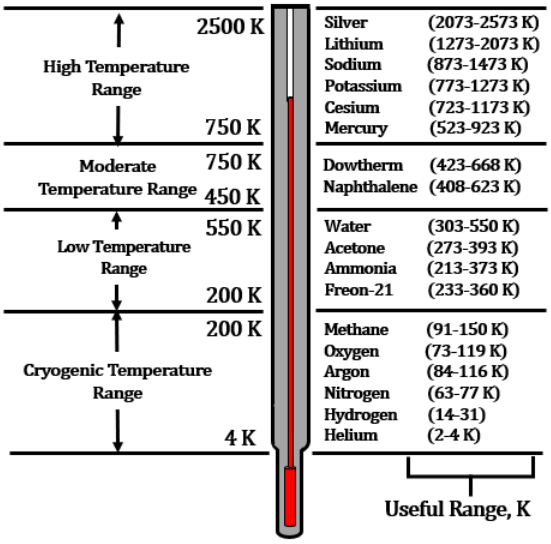


Figure 3. Applicable temperature ranges of common working fluids observed in nuclear

The applicable temperature ranges of low and intermediate-temperature heat pipes are 200 K to 750 K. Ammonia heat pipes have been seen in decay heat removal in the spent fuel pool (Refs. 79, 85, and 121). Water heat pipes, defined as low or intermediate heat pipes, have been observed in a wide range of nuclear applications, including space reactor core cooling (Ref. 8), space reactor heat rejection radiators (Refs. 39 and 97), hybrid heat pipe-control rods (Refs. 61 through 63, 98-99, 123-124), emergency cooling systems (Refs. 56 and 102), electronic reactor cooling (Refs. 100-101), space reactor loop heat pipe radiators (Refs. 42 and 101), microreactor heat exchangers (Refs. 105 through 118, 120), seawater desalination processes (Ref. 126), and nuclear-spent fuel pool heat removal (Refs. 82-83, 122), space reactor heat dissipation (Ref. 127).

The range of high-temperature heat pipes is 750 K to 2500 K. Mercury was considered a working fluid (Refs. 128 and 129) and was initially rejected as a working fluid due to its poor wettability, toxicity, and difficulty in handling (Ref. 130). However, recent mercury heat pipe studies suggest that mercury heat pipes have good thermal stability and low saturated vapor pressure, making them viable for transitioning from water heat pipes to liquid metal heat pipes (Refs. 131-133). Cesium heat pipes were observed in the space reactor's waste heat rejection radiators (Refs. 38, 134) and nuclear reactor cooling (Ref. 96). Potassium heat pipes have been observed in space reactor core cooling (Refs. 25, 38, 135, 93, 94, 107, 136, and 137), heat rejection radiators (Ref. 17), passive residual heat removal for molten salt reactors (Ref. 104), high-temperature waste recovery (Ref. 105), small modular reactor heat exchangers (Ref. 109), microreactor primary cooling mechanism (Ref. 67), and molten salt reactor cooling (Ref. 131). Sodium heat pipes have been observed in space reactor core cooling (Refs. 5, 14, 31 through 33, 36, 50, and 115), space reactor radiators (Ref. 92), hybrid heat pipe-control rods (Ref. 98), passive

heat dissipation of marine nuclear reactors (Ref. 103), heat-pipe nuclear battery type reactors (Ref. 43), microreactor cooling (Ref. 114), passive reactor cooling for heat pipe cooled reactor (Ref. 108), micro molten salt reactor cooling for space or oceanic reactors (Ref. 71), nuclear spaceship applications (Ref. 138), heat-pipe heat exchangers for salt-cooled fission and fusion reactors (Refs. 111 and 139), Stirling hot-side heat exchangers (Ref. 37). Lithium has been seen in split-core heat-pipe reactors (Ref. 13), space reactor core cooling (Ref. 14), H plasma testing (Ref. 76), heat removal from space nuclear reactors (Ref. 113), nuclear thermionic space power system cooling (Ref. 10), heat pipe space nuclear reactor cooling applications (Ref. 16).

II.C. Principles of operation and circulation

Heat pipes successfully transfer large quantities of heat by inducing enough capillary pressure to passively actuate the phase-changing working fluid circulation. The driving pressure is accomplished through a wicking structure for conventional heat pipes, typically firmly bonded to the heat pipe wall. A heat pipe can operate within a specific temperature range and power throughput based on the heat pipe wall, wick structure/material, and working fluid. However, other forces, such as gravitational, centrifugal, electrostatic, and/or osmotic, may be present in applying the heat pipe, either purposefully acting as a supplement or passively acting as a detriment for the passively-driven circulation. As such, heat pipes are susceptible to their environment. It thus becomes imperative to report the environment a heat pipe is exposed to precisely. This section summarizes the principles of a heat pipe operation under startup and normal operational conditions, various heating and cooling methods reported for nuclear-related heat pipe experimental efforts, along with a given heat pipe's operational limitations.

II.C.1. Conventional heat pipe operation

Heat pipes successfully operate when proper heating and cooling methods are employed, which produce stable heat transport rates once the steady state has been reached. However, heat pipes always begin at ambient temperatures, which depending on the intended scenario, may require extra steps to bring the heat pipe to a state at which it can fully function. Most heat pipes observed in nuclear reactors are liquid metals with melting temperatures much higher than ambient temperatures. Before a heat pipe can operate at its full potential, the solid metal inside the heat pipe must be melted to permeate into the wick. This process, known as the startup phase, is integral to the efficacy of the heat pipe. There are essentially three types of startups: uniform, frozen, and gas-loaded startups.

Uniform startup begins the heat pipe at its intended heat transfer rate, often observed in heat pipes whose working fluid begins in liquid. For example, a copper-water heat pipe, where the working fluid existed in the liquid state at the start, does not need to step up the power incrementally since the water has already permeated into the wick.

Frozen startup requires incremental increases in power to melt the working fluid, ensuring that the entire working fluid has changed to a liquid state. Practically all heat pipe studies conducted for nuclear-related technologies practiced the usage of this startup phase. The process begins by conducting enough heat only into the evaporator section to melt the working fluid. Once the working fluid permeates the wick, vapor forms and travel in the evaporator to the adiabatic section. The vapor then condenses in the lower adiabatic section closer to the evaporator section, releasing its latent heat and reverts back to a liquid, which returns to the evaporator section through the wick. This process is repeated until the melt front reaches the condenser end cap. Once the

melt front reaches this point, the axial temperature profile along the length of the heat pipe will be close to uniform.

Gas-loaded startups are similar to frozen startups in that heat is incrementally fed into the heat pipe. The difference between frozen and gas-loaded startups is that some heat pipes contain a non-condensable gas (NCG) evenly distributed in the vapor space, almost always an inert gas that does not mix with other materials. Once the heat is added to the evaporator section, the vaporized working fluid pushes the NCG towards the condenser section. Since gases have much lower thermal conductivity, the presence of the NCG in the condenser section prevents heat from being removed from the system and causes the temperatures of the evaporator and adiabatic sections to increase. Additional heat is added incrementally, continually increasing the vapor pressure and compressing the NCG in the condenser section, moving the vapor-gas front. Once the vapor-gas front is broken, the regular operation can occur.

II.C.2. Methods of heating/cooling heat pipes

Heat pipe experiments performed in a lab setting use alternative heating and cooling methods to simulate the heat pipe's behavior in an actual reactor core. Consequently, if placed in an actual reactor, certain phenomena in the heat pipe are omitted. If heat pipes are operating in an actual reactor core, the neutron bombardment caused by the fission reactions will cause neutron activation in the alkali-metal working fluid. These conditions would cause NCGs to accumulate at the condenser end and gradually decrease the heat rejection rate and power throughput until operational limits are met and the heat pipe efficacy diminishes, ultimately causing heat pipe failure. Space nuclear reactors, in particular, operate under a fast neutron spectrum, which can decrease the structural material's hardness and also cause volumetric expansion, which could weaken the tensile strength of the heat pipe wall, leading to potential leakages of the alkali-metal working fluid along the grain boundaries (Ref. 90). In small-scale heat pipe test facilities, research is conducted with the environmental conditions in mind, seeking to account for discrepancies between its actual intended conditions in the research setting. Especially for space nuclear reactor systems, heat pipe experiments are performed under vacuum conditions to remove natural convection, only allowing radiative heat transfer to simulate the vacuum of space.

The heating source provided to the evaporator section in a heat pipe test facility is usually electrical, provided through radiative heat transfer. However, other methods have used conductive, convective, and/or radiative heat transfer combinations. For nuclear applications, heating has been accomplished through rf-induction coils (Refs. 94, 96, 108, 113, 115, 134, and 140), electric tape (Ref. 97), resistance coil heaters (Refs. 19, 93, 122, 136, 137, 141), cartridge heaters (Refs. 32, 56, 142, and 143), proton/deuteron beams (Ref. 75), hot gas or water loop(s) (Refs. 79-80, 116, 120, and 144), copper electrodes (Refs. 62-63, 98-99, 123-126), electric wire heaters (Refs. 95, 106-107, 109, and 122), electric heater belts (Ref. 82), electric band heaters (Ref. 83), electric furnaces (Refs. 103, 145, and 146), and electric heating rods (Refs. 104 and 148).

The condenser section of heat pipe test facilities related to nuclear technologies is cooled using multiple methods, including water-cooled calorimeters (Refs. 19, 32, 92, 94, 97, 137-141, and 148), forced water cooling (Ref. 56), thermal radiation in vacuum (Refs. 113, 134, 140, and 149), closed helium cycle with compressor cooling machine (Ref. 77), heat exchangers (Refs. 116, 120, 142, and 147), water jackets (Refs. 62, 82-83, 93, 96, 106-107, 122-125, and 144), air tunnel condenser (Ref. 81), ambient environment (Refs. 79, 95, 103, 106-109, 115, 137, 145-150), chiller

calorimeter (Refs. 99 and 143), electric air fan (Refs. 143 and 151), and skutterudite TEG modules coupled with cooling water system (Ref. 105).

II.C.3. Operational heat transfer limitations

The observed steady-state and transient behavior of conventional heat pipes during startup and normal operations are attributed to their design characteristics. Each heat pipe iteration determines the size, shape, working fluid, wick structure/material, and wall structure/material determine the operational limitations (Refs. 1, 91). Operational limits, also known as heat transfer limits, are each described by an axial heat transport range as a function of temperature. The operational conditions imposed on a given heat pipe test facility must produce heat transfer rates within the bounds of its axial heat transport capability. Once an operational limitation has been reached, the heat pipe efficacy begins to diminish and, if continued, will result in heat pipe failure. The most common failures occur when the working fluid's circulation and/or phase-change is weakened or slowed to inhibit the heat transfer, causing hot spots to form. Each operational limit describes a physical phenomenon that may occur under certain operational conditions. The operational limits to increase power throughput and temperature are the viscous, sonic, capillary, entrainment, boiling, and heat rejection limits (Ref. 90).

The *viscous limit* occurs because at low temperatures, the viscous forces are dominant in the flow of the vapor phase, and they may be too large to sustain the movement of the vapor. Mainly, the vapor pressure is lowest at the condenser end and cannot drop below zero, so the viscous pressure drop cannot exceed the evaporator vapor pressure. The viscous limit occurs regularly in the frozen startup of heat pipes, where the initial pressure may be very low; however, this does not lead to a severe failure since the consequence is local temperature rise, which sharply increases the viscous limit. The *sonic limit* occurs because of choking, similar to what may occur in the throat of a converging-diverging nozzle. Along the length of the evaporator from the end cap to the exit, mass is progressively added to the vapor channel, thus yielding a maximum velocity and minimum pressure at the exit. For a given evaporator end cap temperature, as the power increases, this maximum velocity increases until it becomes sonic. At this point, downstream conditions do not propagate upstream of the evaporator exit, so, for example, decreasing the condenser temperature does not increase the rate of heat transfer. The constraint on the velocity that it cannot exceed the sonic speed at the evaporator exit corresponds to a limit on power. The sonic limit is a regular occurrence in the startup of liquid-metal heat pipes. It does not represent a severe failure, as exceeding the sonic limit leads to a temperature increase in the evaporator, which raises the evaporator end cap temperature and thus the sonic limit making it far from isothermal (Refs. 1,91).

The *capillary limit* also called the wicking limit, is encountered when the sum of the pressure drops in the liquid and vapor phases exceeds the maximum capillary pressure in the wick, which implies the wick cannot return liquid to the evaporator fast enough to sustain the operation, leading to dryout. Starting at the evaporator end cap, the vapor undergoes a pressure drop across the length of the heat pipe due to viscous and inertial forces, and the liquid undergoes a pressure drop due to viscous forces when returning to the evaporator. If the heat pipe is not horizontal, a gravity head also affects this limit, either favorably or adversely, corresponding to the condenser above or below the evaporator. The *entrainment limit* occurs because liquid at the wick surface may become entrained in the high-velocity vapor stream without adequate shelter. Waves may develop at the liquid-vapor interface, grow in amplitude, and separate from the surface. This phenomenon is

opposed by surface tension, which draws the liquid back toward the surface. This limit has not been verified experimentally for capillary-driven heat pipes, as the wick surface structure diminishes the development of surface waves and thus may not represent a serious design consideration for the design of capillary-driven heat pipes (Refs. 1,91).

The *boiling limit* occurs because boiling may occur in the evaporator at the wall and/or inside the wick structure under high radial heat fluxes. Some degree of boiling may enhance heat transfer, but if this boiling is severe, the resultant vapor bubbles can become trapped in the capillary structure and impede capillary pressure and liquid flow area, preventing adequate circulation of the working fluid, leading to dryout. This limit is noted to be rarely encountered for liquid-metal heat pipes. The *heat rejection limit* is not necessarily a limitation of the heat pipe itself but rather in a particular environment. The heat transfer through a heat pipe is limited by the rate of heat rejected into its environment. Thus, the boundary conditions of the heat pipe must be considered, which may include a combination of conduction, convection, and radiation. An essential consideration of this limit is the accumulation of non-condensable gases at the end of the condenser, which shortens the effective length of the condenser. These non-condensable gases may be initially present in the heat pipe due to manufacturing flaws or design, and they may also gradually accumulate in the heat pipe due to neutron activation (Refs. 1,91).

Ponanpaan provided data for the sonic, capillary, entrainment, boiling, and condenser limits for a gas-loaded sodium heat pipe as a function of temperature (Ref. 141). Jang provided the sonic and capillary limits for a potassium heat pipe as a function of temperature (Ref. 93). Dickinson provided the viscous, sonic, capillary, and entrainment limits for the homogenous wick heat pipe and the viscous, sonic, and entrainment limits for the annular gap wick heat pipe and the arterial wick heat pipe (Ref. 137). Reid provided the sonic and viscous limit as a function of temperature for the sodium heat pipe in the SAFE-30 reactor prototype (Ref. 32). El-Genk and Tournier provided the sonic, capillary, and entrainment limits for the most extended sodium heat pipe in the SAIRS-A reactor and the potassium heat pipe's sonic, capillary, and entrainment limits in the SAIRS-C radiator (Ref. 36). Dissinger provided the sonic, wicking, viscous, and entrainment limits as a function of temperature for a Titanium/Cesium heat pipe for high temperature heat radiator panels for nuclear-electric propulsion systems (Ref. 134). Kim presented the conventional heat pipe's sonic, capillary, boiling, viscous, and entrainment limits (Ref. 98). Tharayil provided the sonic, capillary, entrainment, and boiling limits for the investigated miniature loop heat pipe for 20%, 30%, and 50% filling ratios (Ref. 100). Park provided the sonic, capillary, viscous, boiling, and entrainment limits for a sodium heat pipe meant for nuclear applications (Ref. 138).

III. HEAT PIPE EXPERIMENTAL RESEARCH REVIEW

Several literature reviews have been reported for heat pipes regarding nuclear technologies. Parker presented various applications of heat pipes in nuclear reactor engineering while also performing feasibility studies to use heat pipes containing liquid fissionable material as control devices for a nuclear reactor (Ref. 152). Ranken presented the application of heat pipes in nuclear reactor space power systems and the characteristics of heat pipes that favor space power systems (Ref. 153). Reid reviewed liquid metal heat pipe work at Los Alamos national laboratory concerning reactor technologies in space and terrestrial environments (Ref. 154). Buden summarized space nuclear reactor power systems, mainly focusing on the SP-100 heat pipe cooled

reactor and its developmental status and evaluation of its performance and scalability (Ref. 135). Reid and Merrigan presented the developmental activity of heat pipe and thermosyphon technologies from 1990-1995, detailing heat pipe implementation in spacecraft development and heat pipe modeling (Ref. 155). Anderson evaluated various working fluids for the temperature range of 450 to 700 K for space nuclear electric power systems to explore deep space, detailing proposed experimental designs that used either heat pipes or loop heat pipes as components in a radiator to reject waste heat (Ref. 130). Bragg-Sitton presented ongoing space nuclear systems development in the United States for ambitious space exploration missions (Ref. 156), Jouhara introduced heat pipes, passive safety systems, and hybrid heat pipes (Ref. 157), and McClure built a database of previous research supporting microreactor designs (Ref. 158). Zohuri introduced heat pipe applications in fission-driven nuclear power plants, including advanced high-temperature reactors, small modular reactors, and pebble bed reactors (Ref. 111). Wang presented opportunities and challenges of micro heat pipe cooled reactor systems with high-efficiency energy conversion units, several space-reactor concept reports, and ideas regarding energy support for oceanic activities and remote districts (Ref. 159). Yan presented micro heat pipe cooled reactor technologies developments with a literature review citing several space reactors and microreactor reports. Various integral components of micro heat pipe reactors are detailed, along with each reactor concept's various component options (Ref. 160). Mueller presented heat pipe modeling and simulation efforts for various nuclear systems, citing several modeling codes from 2021 (Ref. 161).

Most previous reviews may only focus on specific types of heat pipes or for a specific application. Most reviews are backdated and missing recent experimental results reported in the literature. The review article is to extend previous efforts and integrate and provide a summary of previously reported heat pipe experimental efforts and update the review with recently reported results in the literature for heat pipes in all nuclear-related applications, including space power (radiators, core cooling, and electricity production), Microreactors (Emergency core cooling, hybrid controlled rods, core cooling), and other heat pipe involvement in other nuclear-related technology (spent fuel pool).

III.A. Space power technology

Most studies on heat pipe technologies in space reactors focused on the reactor system's heat rejection and thermal control aspects. The spacecraft's electronic system components must be protected from the reactor’s fast neutrons within the reactor itself and from high-energy gammas coming from space. Space reactor radiators that employ heat pipes remove unwanted heat from the reactor core and radiate it out into space using heat pipe radiator panels with fins. The most commonly used working fluids for space reactor’s heat pipe systems are alkali-metals, containing and transmitting enormous amounts of energy. Water heat pipe radiators have only been considered for deployable heat pipe radiators after thermionic conversion. At the end of this section, a table (Table I) lists references related to the application of heat pipes in space power technology presented herein.

III.A.1. Space nuclear radiators

The waste heat generated during the thermal to electric power conversion cycle process must be dissipated to avoid overheating the reactor system, easily damaging electronic equipment, and much more fragile spacecraft components. Thus, all space radiators are required to have exposed and highly emissive surfaces to the vacuum of space. Due to the harsh coldness of space

and the cyclical usage of the reactor system, heat pipe space nuclear radiators are required to successfully operate under thermal cycling loads, where the heat pipe starts from the frozen state, heats up to a fully operating heat pipe at steady state, operates under isothermal conditions, cools back down to frozen state, and then repeats the startup and cooldown processes. To further enhance the radiative heat transfer into space, the heat pipe radiator is impeded with fins, which increase the radiative surface area of the exposed section of the heat pipe to space. Originally metallic or composite fins were considered, but due to a large amount of mass added to the spacecraft, concepts have pivoted to using woven bare carbon fiber fins without matrix materials. Using carbon fiber (C-C) fins without matrix materials reduces the radiator mass while preventing a decrease in lower thermal conductivity caused by the usage of matrix material (Ref. 162).

There are essentially two categories of space radiators: rejection at temperatures below 350 K and nuclear or high-power systems at temperatures around 500 K and beyond (Ref. 163). Most direct space reactor cooling using heat pipes is accomplished by coupling heat pipes directly to thermionic converters. Each thermionic converter produces an electrical power output by emitting electrons over a potential energy barrier from a hot electrode to a colder electrode. The evaporator section of each heat pipe is inserted into the reactor core, while the condenser section is coupled to the hot electrode of each thermionic converter.

Deployable heat pipe radiator panels are located far from the reactor core, containing a working fluid that operates at relatively lower temperatures than the auxiliary reactor heat pipes. El-Genk and Huang investigated the transient response of water heat pipes to step changes in input power at different cooling rates on behalf of the Institute for Space Nuclear Power Studies (Ref. 97). The experiment utilized a horizontal copper heat pipe and a double-layered copper screen wick. Uniformly heating using electric tape attached to the outside of the heat pipe wall was applied to the evaporator and a water-cooling jacket to the condenser section. The transient axial distribution of vapor and wall temperatures with a cooling jacket's water mass flow rate of 3.2 g/s and heater power input value of Q_{in} = 570 W is shown in Fig. 4.

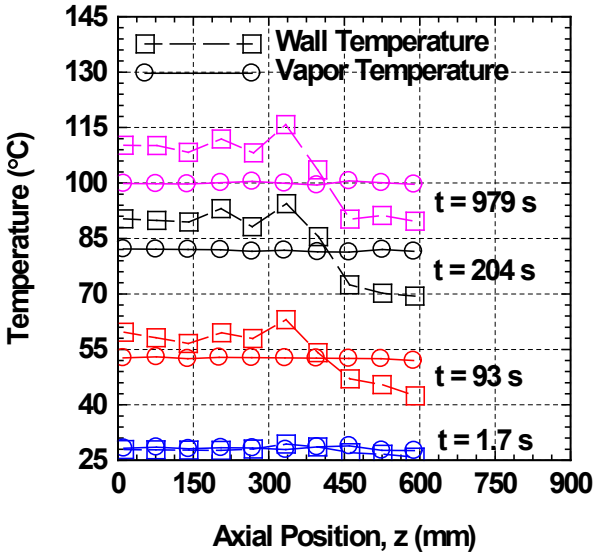


Figure 4. Transient axial wall temperature profile for water heat pipe test (Ref. 97).

Results of the experiment found that an exponential function can describe the heat-up and cool-down transients. The steady-state values and their rates of change at the beginning of the

transient define the time constants for the vapor temperature and the adequate power. The steady-state vapor temperature increases by increasing the power input or decreasing the water flow rate.

Several examples of potassium heat pipe radiators with experimental data sets have been reported. Woloshun experimentally investigated stainless-steel potassium heat pipes for nuclear power system radiators (Ref. 136). Data was provided for the tested heat pipe with stainless steel screen mesh wick. Preliminary ground tests were performed using heat pipes in a vacuum enclosure with electronic resistance heaters. Each heat pipe was placed in a horizontal orientation to mitigate gravitational effects. The evaporator section was heated using a DC heater with a heat load of 412 W was held constant for the test duration. The condenser section was monitored continuously using a water-cooled calorimeter. The calorimeter power asymptotically approached 290 W, indicating significant heat loss through the heater insulation. The transient axial wall temperature profile for the homogenous wick heat pipe with Q_{in} = 412 W is provided in Fig. 5.

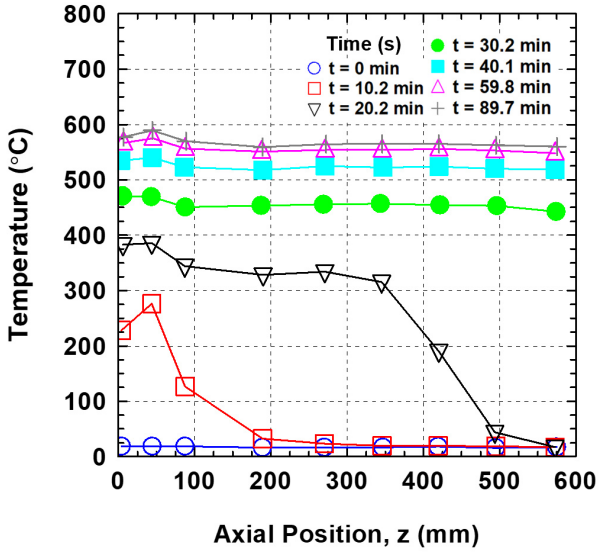


Figure 5. Transient wall temperature response results for stainless-steel potassium heat pipes with Qin = 412 W (Ref. 136).

Jang experimentally and analytically studied the startup characteristics of a potassium heat pipe in a space power heat rejection system (Ref. 93). The stainless-steel heat pipe (457.2-mm in length with a 25.4-mm OD and 1.65-mm wall thickness) was tested in a vacuum chamber. The 1.016-mm thick wick structure was a stainless-steel screen mesh with an approximate porosity of 0.67. The evaporator section was heated using a silicon carbide heater surrounded by a 17-layer radiation shield to reduce heat losses, and the condenser section was cooled using a copper cooling shroud with a 6.35-mm diameter copper tube wrapped around the condenser to circulate the coolant. The transient axial wall temperature distribution for t = 0, 40, and 120 min are reported for heater power input levels of Q_{in} = 500 W and 1700 W in Fig. 6. It was concluded that more heat inputs result in faster heat pipe startups during the startup period but should be maintained at less than the capillary limit at the steady state.

Dickinson investigated stainless steel potassium heat pipes under microgravity conditions for high-temperature thermal control systems such as nuclear-powered spacecraft (Ref. 137). The test apparatus was placed in a horizontal position to minimize gravity effects. The test was conducted at vacuum pressure, and the stainless steel 304 heat pipe was tested with three wick

structures: homogenous screen mesh, annular gap, and arterial wick. Steady-state temperature profiles at various power levels (80-280 W) were reported for the homogenous, annular gap, and arterial wick heat pipes. The homogenous wick wall temperature results are shown in Fig. 7.

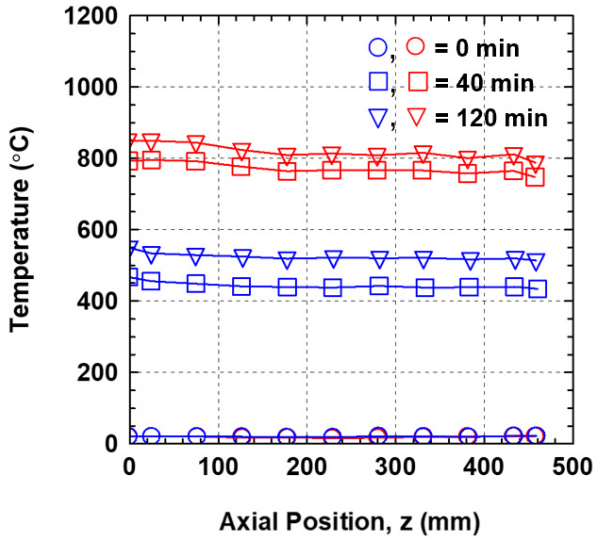


Figure 6. Transient axial wall temperature distributions during startup period with constant heat power input of 500 W (blue) and 1700 W (red) for potassium heat pipe (Ref. 93).

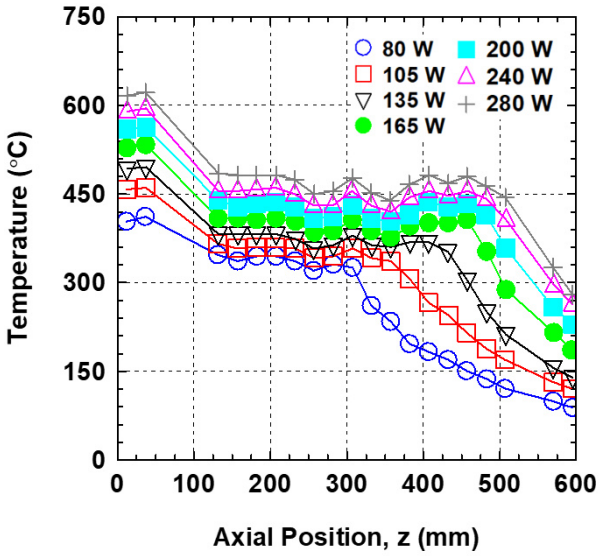


Figure 7. Steady-state axial wall temperature profiles at various heating power input values for high-temperature stainless steel potassium heat pipes under microgravity conditions (Ref. 137).

Glass investigated the startup of an Nb-1%Zr potassium heat pipe from the frozen state on behalf of NASA (Ref. 94). The heat pipe was tested horizontally to minimize gravitational effects. The 3.8-mm thick wick structure consisted of three 100-mesh plain, square weave Nb-1%Zr screen layers. The heat pipe was heated using an induction coil wrapped four times around the 100-mm long evaporator section. A single-pass water calorimeter monitored the cooling mechanism. Steady-state axial wall temperature distribution corresponding to heating input power levels (92.52-601.54 W) is presented, as shown in Fig. 8.

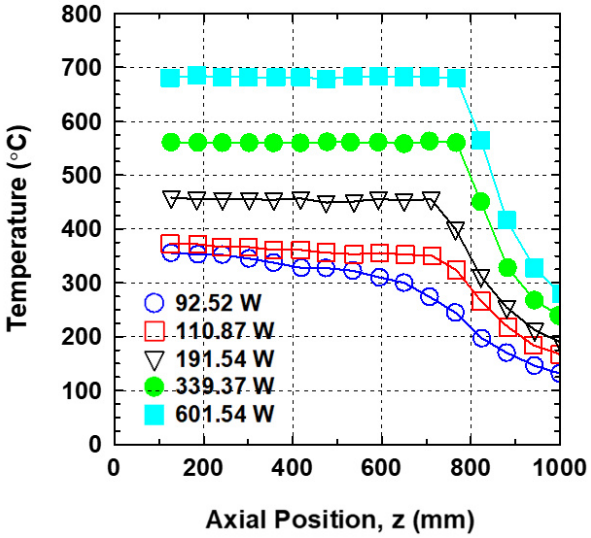


Figure 8. Steady-state axial wall temperature profiles at various power input values for Nb-1%Zr potassium heat pipes from the frozen state (Ref. 94).

Ponnappan studied startup transients and the performance of a gas-loaded sodium heat pipe for large-area space radiators (Ref. 141). The heat pipe was performed under two conditions, namely vacuum conditions and gas-loaded conditions. The non-condensable gas loaded into stainless steel 304 heat pipe was argon. The wick structure was a double-wall stainless steel artery. The heat pipe was heated using electrical resistance coil nichrome heaters wrapped around the evaporator. The cooling mechanism consisted of a circulating chilled water loop from a constant temperature bath through a copper coil wrapped around the chamber. In Fig. 9, steady-state axial wall temperature profiles for the vacuum and gas-loaded conditions are presented for 306-757 W and 338-1223 W, respectively. The gas-loaded liquid-metal heat pipes quickly started from the frozen state without adverse effects, even if with a long adiabatic channel.

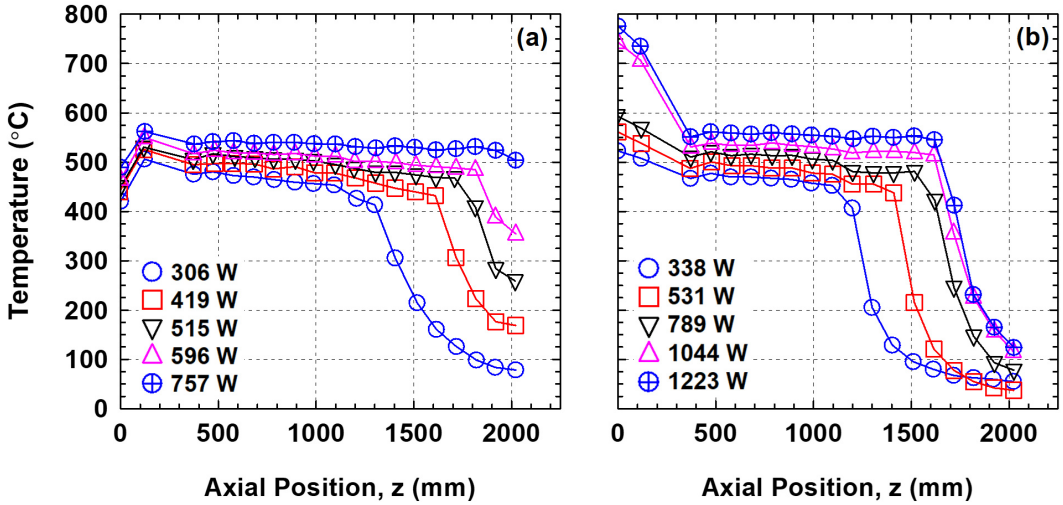


Figure 9. Steady-state test axial wall temperature profiles in (a) vacuum mode and (b) gas-filled mode of stainless-steel sodium arterial heat pipe (Ref. 141).

Kang experimentally investigated the transient behavior of an Inconel 600 sodium heat pipe for space nuclear power system radiator systems (Ref. 146). The heat pipe was tested at

various inclination angles (-15°, 0°, +45°, and +90°) with a heater power input value of 790 W. The heat pipe was 420-mm long with a 13-mm OD and 2-mm wall thickness. The screen wick consisted of one layer of stainless-steel 316 100-mesh. The filling ratio of sodium in the hat pipe was 10% (2.86 g). The 240-mm evaporator section was heated by inserting the heat pipe into the central part of the furnace. The 120-mm condenser section was cooled by the ambient environmental temperature of the room (27°C). The transient axial wall temperature profile along the heat pipe wall during the frozen startup period is shown in Fig. 10. Additional helpful information regarding the temperature as a function of time was also provided when 0% sodium-filled in the empty tube, 10% sodium-filled in thermosyphon without wick, and 10% sodium-filled with screen wick. Transient temperature profiles were also provided for various inclination angles (+90°, 0°, +45°, and -15°). The lowest thermal resistance value was 0.11 °C/W at a 90° angle.

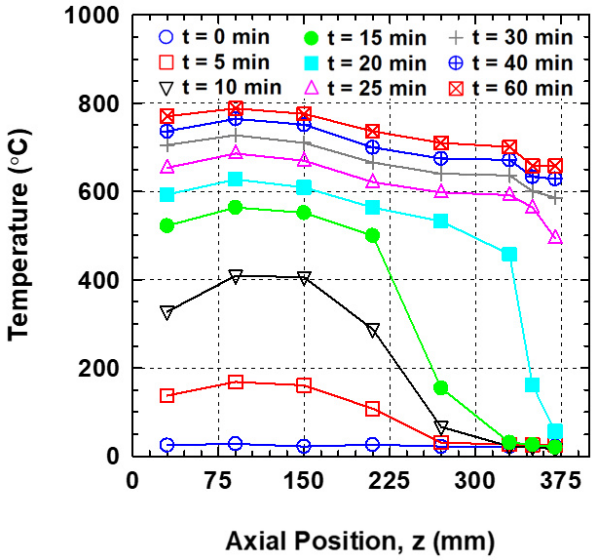


Figure 10. Transient axial wall temperature profiles during a frozen startup period of Inconel 600 sodium heat pipe (Ref. 146).

Ma experimentally investigated the effect of inclination angle on the startup of a frozen sodium heat pipe for nuclear reactor power and residual heat removal systems (Ref. 95). The stainless steel 316 heat pipe and a thick wick consisted of two layers of stainless steel 100/400 mesh. The filling ratio (liquid to conduit volume ratio) of sodium in the heat pipe was 20%. The evaporator section was heated using heating resistance wires wound around the evaporator surface. The voltage controlled the heating power, with the potential heat transfer rates being 0-3500 W. The condenser section was cooled by air convection and radiation at room temperature (about 25°C). The evaporator and adiabatic sections were wrapped in 95-mm thick aluminum-silicate ceramic-fiber insulation. Acrylic aluminum foil tape was then wrapped on the outside. Steady-state wall temperature profiles at various heat input power levels were reported for the heat pipe with a 0° inclination, as shown in Fig. 11. The determined critical startup powers for various inclination angles were provided. At negative inclination angles, the gravity was concluded to hinder liquid flow from the condenser to the evaporator, causing more significant end-to-end temperature differences and greater startup power than in horizontal and positive cases.

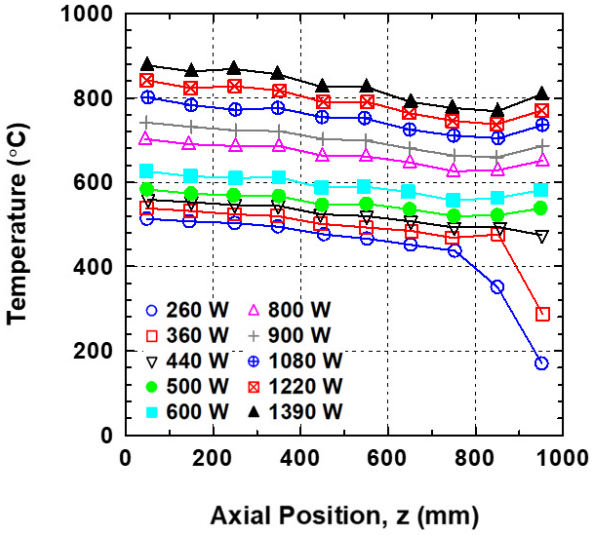


Figure 11. Steady-state axial wall temperature profile at various heat input power levels with a 0° inclination angle for a sodium heat pipe (Ref. 95).

Lithium heat pipes have also been considered for space radiators. Tournier and El-Genk presented thermocouple data results of a lithium heat pipe for a space nuclear reactor's heat rejection radiator system in line with validating the heat pipe code HPTAM (Ref. 113). The molybdenum heat pipe structure was 1800-mm long with a 19.1-mm OD and 1.52-mm wall thickness. A 0.37-mm annular gap separated the mesh screen wick from the heat pipe wall. The wick structure consisted of seven layers of Mo-41w%Re 400-mesh screen. The Li/Mo heat pipe was inserted into a clear quartz (SiO_2) tube. The 300-mm evaporator section was heated using an induction-heating coil outside the quartz tube. The thermal radiation cooled the 1470-mm condenser section to the evacuated quartz tube surrounding the heat pipe. The transient axial temperature profiles are reported for a net evaporator input heat flux of 22.2 W/cm^2 in Fig. 12. The results generated from the code were in good agreement with the experimental data.

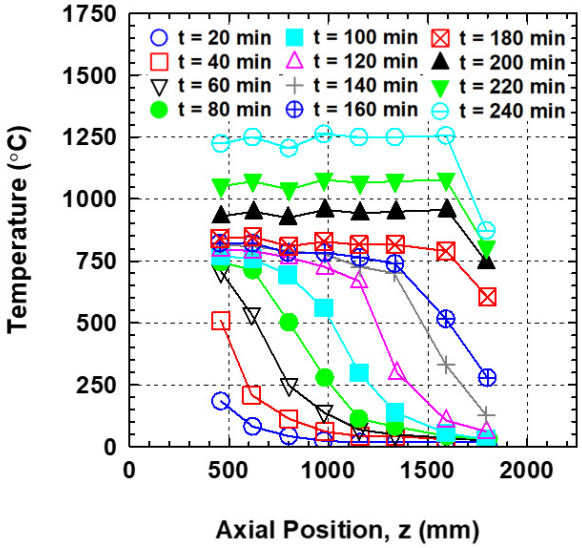


Figure 12. Axial wall temperature profile of lithium heat pipe from the frozen state (Ref. 113).

Chen investigated a cesium heat pipe's frozen startup and heat transfer characteristics under horizontal conditions for nuclear reactor cooling applications, including space radiators (Ref. 96). The Inconel 600 heat pipe with a wick consisted of two metal screens (400 mesh, 100 mesh) layers. The filling ratio of cesium in the heat pipe was 8.8% (48 g). The 180-mm evaporator section was heated using a high-frequency induction heater, and the condenser section was cooled using cooling water circulating loop. Steady-state wall temperature profiles at various heat input powers levels ($772.5 < Q_{in} < 1081.3$) were reported for constant heating ($772.5 < Q_{in} < 1081.3$ W) and a frozen startup process ($772.5 < Q_{in} < 1518.4$), as shown in Fig. 13. The best temperature uniformity was achieved when the filling ratio was 8.8% and the heat power of 940.9 W. A modulated heating mode was recommended to avoid reaching the capillary limit, keep good temperature uniformity, and increase the operating temperature and effective length.

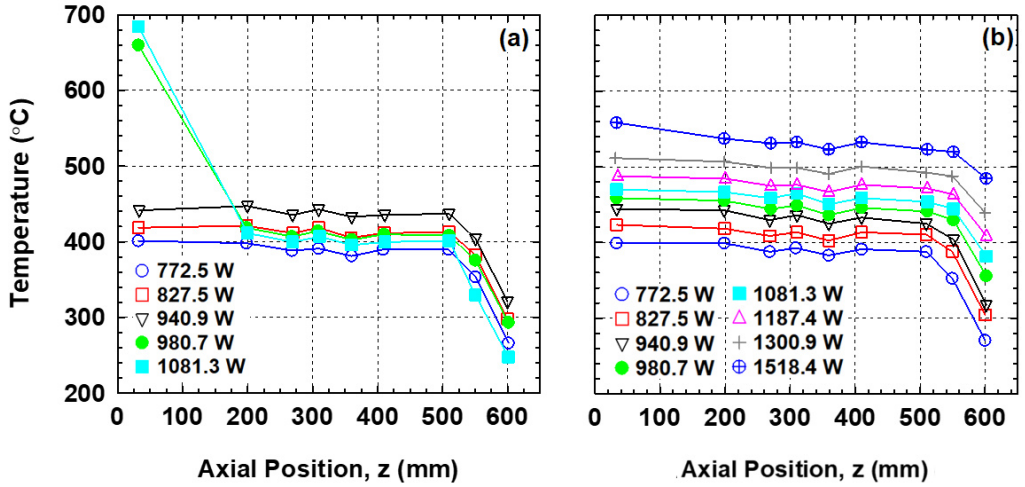


Figure 13. Steady-state axial wall temperature profiles for (a) constant heating and (b) frozen startup process for cesium heat pipe (Ref. 96).

III.A.2. Space nuclear reactor cooling

Tian experimentally investigated the heat transfer performance of high-temperature potassium heat pipes at various inclination angles and heating power input levels for heat pipe cooled reactor systems (Ref. 107). A stainless steel 316 heat pipe with a thick wick structure consisted of stainless steel 316 mesh screen with a porosity of approximately 0.55. Two identical heat pipes were created and tested, with two different heat pipes, 1 containing 20 g (20% filling ratio) and 2 containing 100 g (100% filling ratio). Additionally, two experimental cooling systems (system A and system B) were created, where the condenser section was cooled under air-cooled (system A) and water-cooled conditions (system B). The effects of the heating power (200-4000 W) and inclination angle ($-15° < \theta < 90°$) on the heat transfer performance were reported. The axial wall temperature profiles at a 75° inclination angle are provided for various heat pipe input values for heat pipe 1 (264-580 W) and heat pipe 2 (2164-4115 W) in Fig. 14. The thermal performance of the heat pipe was shown to increase with the inclination angle within 30° and decrease beyond 30°. However, further increase in inclination beyond the critical angle was considered favorable for liquid reflux accelerated by gravity and reduction in thermal resistance.

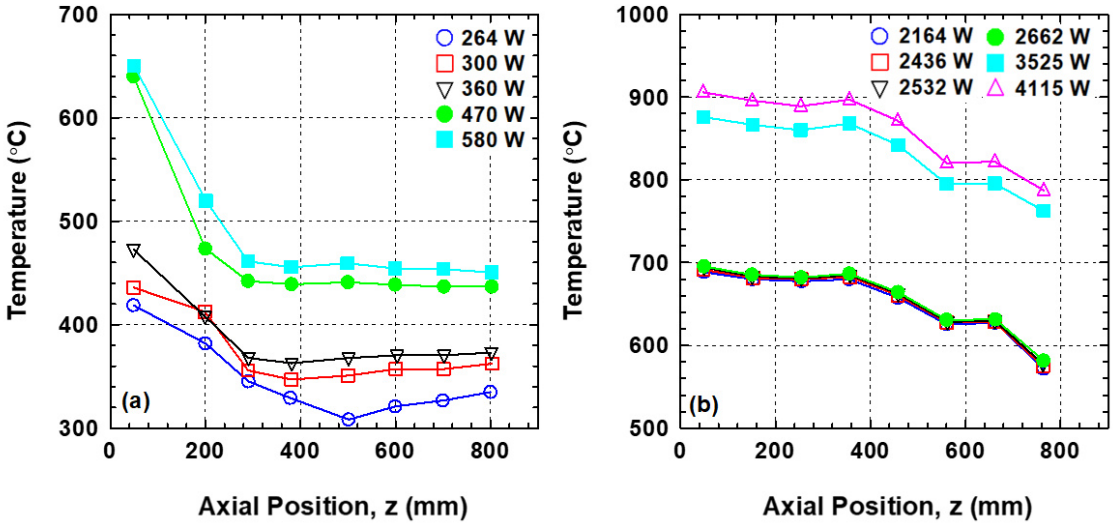


Figure 14. Axial wall temperature profile of heat pipes (a) 1 and (b) 2 at 75° (Ref. 107).

III.A.3. Space nuclear electricity production

Anderson and Tarau presented thermocouple data results for a NaK heat pipe for a Stirling radioisotope system in which electricity is produced using one or more General Purpose Heat Sources (GPHS) units (Ref. 164). In each GPHS, heat is produced using plutonium-238 encapsulated in an iridium cladding surrounded by a graphite insulator. The stainless-steel heat pipe was fitted with a wick structure of a stainless-steel mesh screen. The heat pipe was filled with argon, non-condensable gas. The evaporator section was heated using an RF coil, and the test facility contained two condenser sections cooled by radiation and natural convection. Several steady-state axial temperature profiles were provided, with data sets for various reservoir temperatures (40 and 120°C) and for when the Stirling Converters turned on and off, as shown in Fig. 15. The experiment results demonstrated the feasibility of using a variable conductance heat pipe for backup cooling of the GPHS to allow the Stirling convertor to be stopped and restarted.

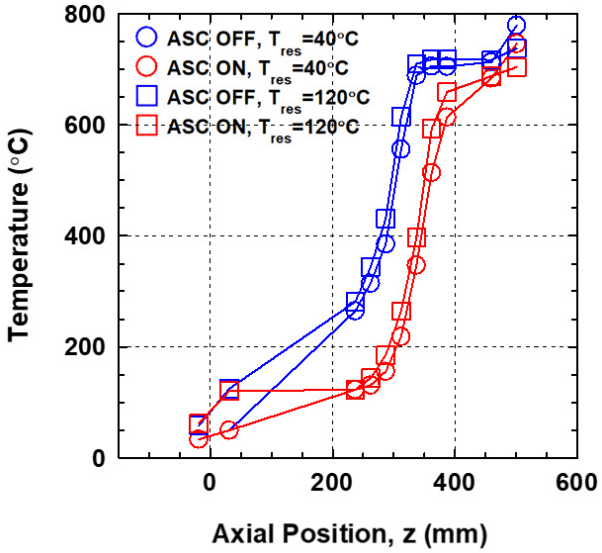


Figure 15. Axial wall temperature under varying conductance NaK heat pipe coupled to a Stirling radioisotope system (Ref. 164).

Table I. Studies of heat pipes in space reactor systems with experimental data sets.

Reference	Working Fluid	Operating Temperature	Wall Material	Wick Design	Dimensions	Heater Power (W)
Ponnappan, 1989 (Ref. 141)	Sodium	800°C	Stainless steel 304	Double-wall stainless steel artery	D_o = 22.2 mm D_i = 18.9 mm L_t = 2000 mm L_e = 250 mm L_c = 910 mm	308-805 (Vacuum) 338-1223 (Gas-Loaded)
El-Genk and Huang, 1993 (Ref. 97)	Water	295-390 K (33-117°C)	Copper	Double-layered 150-mesh copper screen mesh	D_o = 19.1 mm D_i = 17.3 mm L_t = 610 mm L_e = 393 mm L_c = 170 mm	365-590
Woloshun, 1993 (Ref. 136)	Potassium	575°C	Stainless steel	11 layers of 100-mesh stainless steel screen mesh	D_o = 23.0 mm D_i = 21.22 mm L_t = 610 mm L_e = 100 mm L_c = 460 mm	412
Jang, 1995 (Ref. 93)	Potassium	700-1200 K	Stainless steel	Stainless steel screen mesh	D_o = 25.4 mm D_i = 22.1 mm L_t = 457.2 mm L_e = 139 mm L_c = 304.8 mm	285.1-1250
Dickinson, 1996 (Ref. 137)	Potassium	600°C	Stainless Steel 304	11 layers stainless steel 304L screen mesh	D_o = 23.0 mm D_i = 21.22 mm L_t = 610 mm L_e = 89 mm L_c = 521 mm	80-280
Glass, 1998 (Ref. 94)	Potassium	675°C	Nb-1%Zr	Three layers of Nb-1%Zr plain, square-weave 100-mesh screen with 5.1-mm annular gap	D_o = 7.75 mm D_i = 6.48 mm L_t = 1000 mm L_e = 100 mm L_c = 807 mm	92.52-601.54

Reference	Working Fluid	Operating Temperature	Wall Material	Wick Design	Dimensions	Heater Power (W)
Tournier and El-Genk, 2001 (Ref. 113)	Lithium	1500 K (1227°C)	Molybdenum	Seven layers of Mo-41w%Re 400-mesh screen with 0.37 mm annulus	D_o = 19.1 mm D_i = 16.06 mm L_t = 1800 mm L_e = 300 mm L_c = 1470 mm	4000
Anderson and Tarau, 2008 (Ref. 164)	NaK	700°C	Stainless steel 304L	Stainless steel 304L mesh screen	D_o = 19.1 mm L_t = 600 mm L_e = 114 mm $L_{c,1}$ = 73 mm $L_{c,1}$ = 66 mm $L_{c,inactive}$ = 50 mm	250
Kang, 2019 (Ref. 146)	Sodium	775°C	Inconel 600	One layer of Stainless-steel 316 100 mesh screen	D_o = 13 mm D_i = 9 mm L_t = 420 mm L_e = 240 mm L_c = 120 mm	790
Ma, 2021 (Ref. 95)	Sodium	550-900°C	Stainless steel 316	Double-layer 100/400 stainless steel mesh screen	D_o = 20 mm D_i = 16 mm L_t = 1000 mm L_e = 400 mm L_c = 400 mm	260-1390
Tian, 2021 (Ref. 107)	Potassium	550-900°C	Stainless steel 316	Stainless steel 316 300-mesh screen	D_o = 30 mm D_i = 23 mm L_t = 800 mm L_e = 230 mm L_c = 370 mm	200-800 (System A) 800-4000 (System B)
Chen, 2022 (Ref. 96)	Cesium	500°C	In600	Two layers of metal screen meshes, (400 mesh, 100 mesh)	D_o = 32 mm D_i = 25 mm L_t = 600 mm L_e = 180 mm L_c = 420 mm	940.9

Subscripts: t: total, e: evaporator, o: outer, i: inner, and c: condenser

III.B. Microreactor power

The heat pipes incorporated into microreactors, and small modular reactors have a large diversity in application. Heat pipes have been implemented in emergency core cooling systems, hybrid heat pipe control rods, and direct cooling of the reactor. At the end of this section, a table (Table II) lists references related to the application of heat pipes in microreactors power technology presented herein.

III.B.1. Emergency core cooling systems (ECCS)

The emergency core cooling system (ECCS), a vital component of the reactor safety system, is required for all reactor concepts. Heat pipes have been proposed as the emergency heat transfer mechanism during accident scenarios. Seo investigated the length effect on large-L/D vertical heat pipes (thermosyphons) entrainment limit in nuclear power plant ECCS during station black-out scenarios due to beyond design-basis accidents (Ref. 118). The working fluid was distilled water, and the thermosyphon filling ratio was 100% (0.216 L). The stainless-steel heat pipe had an evaporator impeded with 24 cartridge heaters. The condenser section was cooled using a water jacket. The evaporator and adiabatic sections were covered in fiberglass insulation, followed by an outer layer of aluminum foil to mitigate radiation heat transfer. The steady-state axial wall temperature profile for 600-2100 W power was reported in Fig. 16. The heat-pipe performance exhibited similar trends for various power levels but differing values. Additionally, the entrainment limit for the heat pipe differed significantly from predictions of conventional correlations.

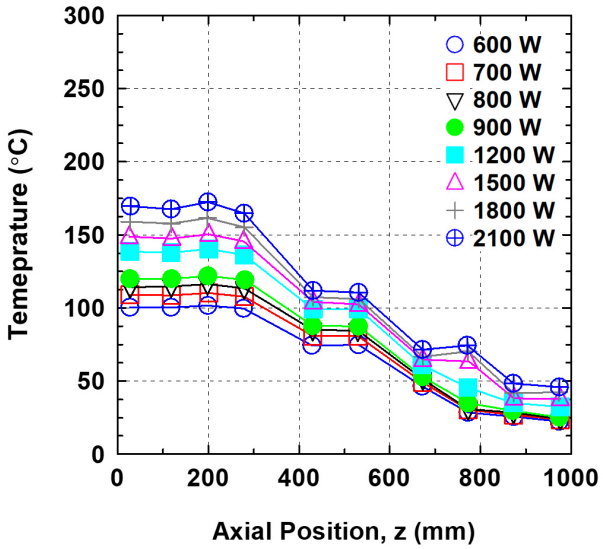


Figure 16. The axial wall temperature profile of wall surface temperature of thermosyphon at various positions (Ref. 118).

Teng experimentally investigated a sodium heat pipe's start-up and thermal performance behavior under swing conditions for passive heat dissipation technology for marine nuclear reactors in an emergency (Ref. 103). The stainless steel 310 heat pipe utilized a double-layer stainless steel 20-mesh wire screen. The evaporator section was heated using an electric furnace with four silicon carbide rods. The condenser section was cooled directly by natural air convection. A steady-state axial wall temperature profile was provided for the heat pipe in a vertical orientation

($\alpha = 0$) without any swing ($\omega = 0$), as shown in Fig. 17. The swinging motion was shown to have a negligible effect on the whole start-up performance of the heat pipe. However, small-amplitude periodic temperature fluctuations were observed, especially in the evaporator section. The temperature fluctuation frequency and amplitude observed in the heat pipe corresponded with the swing frequency and amplitude.

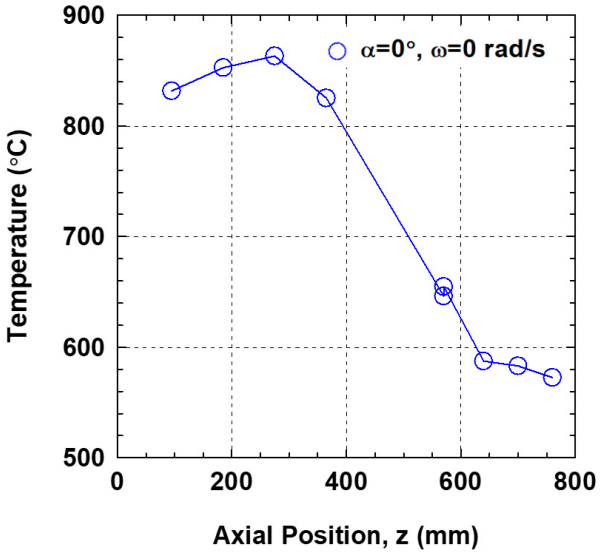


Figure 17. The steady-state wall temperature profile of sodium heat pipe at a static state (Ref. 103).

III.B.2. Hybrid heat pipe control rods:

The pressurized hybrid heat pipe control rod concept has been proposed for advanced nuclear power plants as a passive heat transfer device. The hybrid heat pipe concept is a water thermosyphon with a neutron absorber composed of B_4C pellets held together by a layer of metal cladding and a holding spring. The neutron absorber in the metal cladding is located only in the evaporator section, the portion exposed to the inside of the reactor core. By inserting a neutron absorber into the evaporator section, the hybrid heat pipe achieves both the functions of a heat pipe and a control rod, simultaneously removing decay heat generated from the reactor core while having the ability to shut down the reactor under accident scenarios. Many small modular reactor concepts have considered using hybrid heat pipe control rods instead of regular ones. The NuScale SMR concept contains a hybrid heat pipe as its Passive IN-core Cooling system (PINCs), where the evaporator section is inserted in the reactor core, with a control rod drive mechanism enabled to control the reactivity and safe shutdown of the nuclear fission reaction by inserting the neutron absorber system. The condenser section is cooled by a liquid pool located outside the reactor pressure vessel, where the stored heat of the heat sink can be transferred to the environment.

Kim investigated the heat transfer characteristics and operation limit of pressurized hybrid heat pipes for small modular reactors such as the NuScale SMR (Ref. 123). The stainless steel 316L heat pipe was wickless. A long rod consisting of B_4C neutron absorber pellets is placed at the center of the heat pipe in the evaporator section and is fixed in place by a spring. The heat pipe was heated using two copper electrodes connected to a power supply placed at the top and bottom of the evaporator section. The condenser section was cooled using a water jacket. All sections of the heat pipe were wrapped in thick fiberglass insulation. Axial temperature profiles for 0.2, 3, 5,

10, 15, and 20 bar of pressure were reported for heating power values varying from 200-5800 W. Axial wall temperature profiles for 600 W are provided in Fig. 18. The effects of operating pressure on the evaporation heat transfer were considered negligible. The condensation heat transfer was affected by the amount of non-condensable gas in the test section.

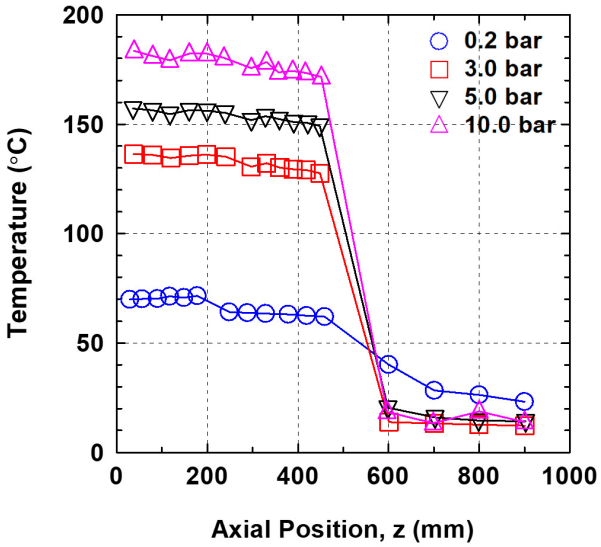


Figure 18. Axial temperature profile for Qin = 600 W at various operating pressures for pressurized hybrid heat pipe-control rod (Ref. 123).

Kim investigated the heat transfer performance of annular thermosyphon heat pipes for Passive In-Core Cooling systems (PINCs) for nuclear safety systems, such as the hybrid heat pipe control rod (Ref. 62). Experimental data sets were reported for four different heat pipes: thermosyphon (THP), annular thermosyphon (ATHP), capillary-driven heat pipe (CHP), and annular capillary-driven heat pipe (ACHP). The working fluids observed were distilled water and FC-72. The stainless steel 316L heat pipe was fitted with a single-layer mesh screen wick. The evaporator section was heated using direct-current copper electrodes. The condenser section was heated using a water jacket. Axial wall temperature profiles for THP, ATHP, CHP, and ACHP were presented for various heating power input values between 30 and 130 W. The axial temperature profile for all investigated heat pipes with Q_{in} = 90 W is shown in Fig. 19. The shear viscosity force was dominant compared to the buoyancy force in the annular thermosyphon.

III.B.3. Heat pipe cooled reactors

Heat pipe cooled reactors utilize heat pipes to cool the reactor core and transfer the fission heat passively. Some heat pipe cooled reactor concepts generate electricity directly by sandwiching high temperature skutterudite thermoelectric generators (TEGs) between the heat pipe condenser sections and cooling water plates. Heat pipe TEGs are coupled to the nuclear reactor by placing each unit in between layers of fuel rods. Heat pipes can also remove decay heat from dump tanks of molten salt breeder reactors and heat removal from the first wall of fusion reactors.

Panda compared numerical and experimental results of a high-temperature sodium heat pipe for passive heat removal in the ^{233}U-Thorium fueled Compact High-Temperature Reactor (CHTR) (Ref. 150). The CHTR contains a set of sodium heat pipes inserted in the upper plenum of the reactor to passively transfer heat to the heat utilization vessels located above the upper

plenum. An additional set of heat pipes are also present to transfer heat from the upper plenum to the atmospheric air during postulated accident conditions (Ref. 165).

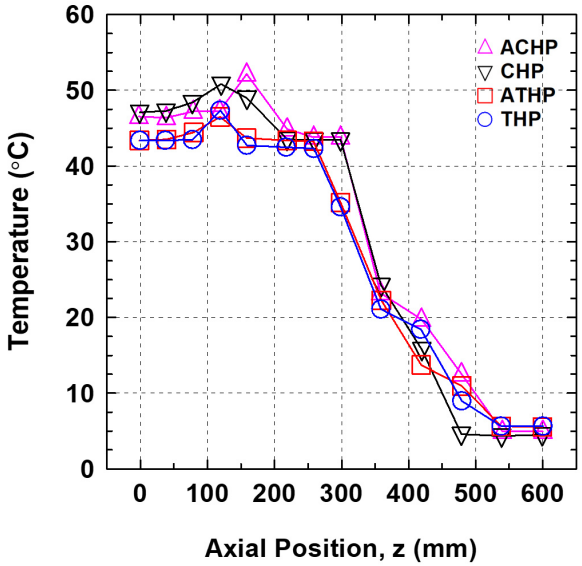


Figure 19. The axial wall temperature profile of THP, ATHP, CHP, and ACHP for 90 W (Ref. 62).

The cross-section of the tested heat pipe was a square. The orientation of the heat pipe setup was vertical. The stainless steel 304 heat pipe used a wick structure consisting of four stainless-steel mesh wire screens. The heat pipe was filled with molten sodium (~120% of wick volume). The heat pipe was tested with 40 kW/m^2 heat flux input, equivalent to 3815.15 W. The experimental data's steady-state axial wall temperature profile and axial velocity distribution of the working fluid are provided in Fig. 20. The transient wall temperature profile as a function of time was also provided.

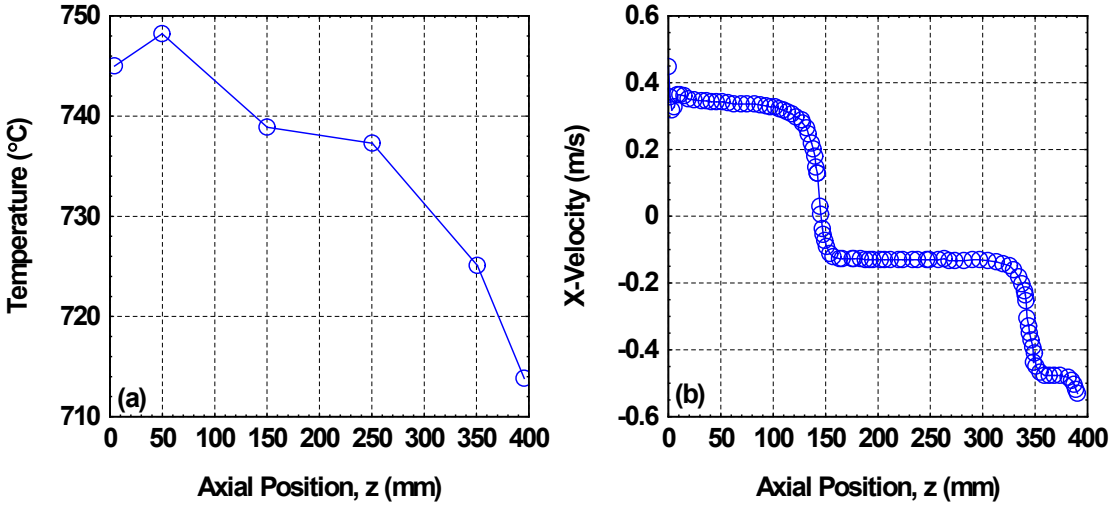


Figure 20. (a) Axial wall temperature distribution of sodium heat pipe at steady state, (b) Longitudinal distribution of the x-velocity component along the axial length of the heat pipe (Ref. 150).

Tang investigated the thermoelectric performance study of a heat pipe thermoelectric generator for micro nuclear reactor applications (Ref. 147). The potassium heat pipe was 800-mm long with a 28-mm OD and 3-mm wall thickness. The material specification of the wick and wall structures were not mentioned. The wick structure was specified as 300-mesh. The evaporator section was heated using six electrical heating rods placed in a copper block where the heat pipe evaporator section also resides. The condenser section was also fitted in a separate copper block, sandwiched between layers of skutterudite TEGs and cooling water plates. The entire experimental setup was sealed in a vacuum chamber. Data were reported comparing the heat pipe wall temperature, Cu-Eva temperature, Cu-Con temperature, and TEG cold side temperature experimental data at steady-state with simulation results generated from the FEM software COMSOL Multiphysics. The heat pipe wall temperature is shown in Fig. 21. Results showed that the influence of the thermal resistance on the TEG hot side was more significant than on the TEG cold side. When the heat flux was increased, the temperature difference of the TEG module and the p-n unit increased along with the thermoelectric conversion efficiency improving.

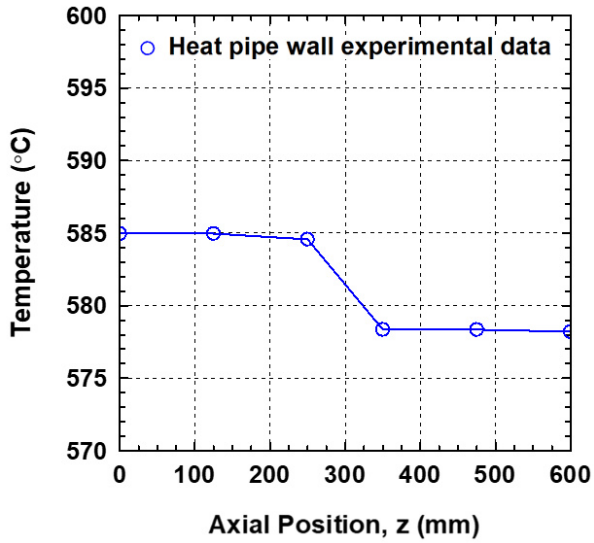


Figure 21. The axial steady-state wall temperature profile of coupled heat pipe to a thermoelectric generator (Ref. 147).

Wang proposed a novel high-temperature waste heat recovery system with heat pipes and a thermoelectric generator (Ref. 105). The stainless-steel heat pipe was filled with sodium with a stainless-steel 300-mesh screen wick. The evaporator section was inserted into a copper block and was heated using electric heating rods. The condenser section was cooled by inserting the condenser section into a copper block with skutterudite TEG modules sandwiched between the copper block and heat exchangers for electricity generation. Steady-state and transient axial wall temperature profiles are provided for different cooling methods using air and water, as shown in Fig. 22. The cooling conditions were shown to impact the heat transfer performance significantly. The isothermality of the heat pipe in the TEG was more significant than that under direct cooling conditions (air and water cooling). The effective thermal conductivity increased with operational temperature, showing that when the heat pipe operation temperature reached 630°C, the heat pipe's effective thermal conductivity reached 35831 W/m·K, almost 100 times that of copper.

Ji investigated the thermal performance of high temperature sodium heat pipes for heat pipe cooled nuclear reactors (Ref. 108). Several heat pipes were investigated with different heat

power, heating positions, and lengths of evaporator section. The stainless steel 310S heat pipe with a wick structure consisted of two outer layers of 400-mesh and two middle layers of 200-mesh. The evaporator section was heated using an electromagnetic induction high-frequency heater. The condenser section was cooled using a copper coil water-cooled jacket. An axial wall temperature profile in a vertical orientation is shown in Fig. 23.

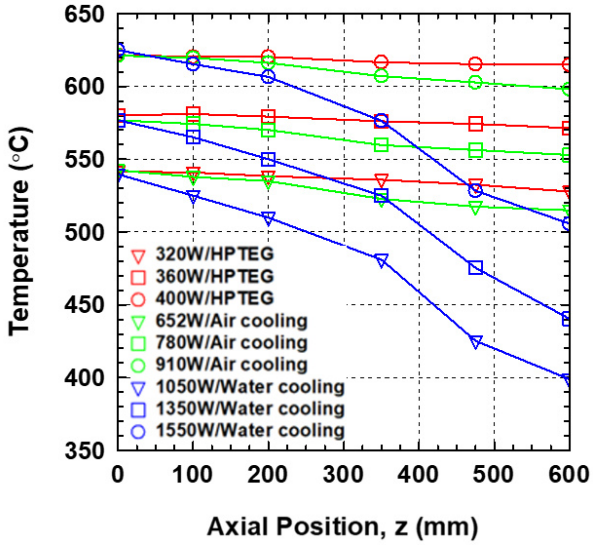


Figure 22. Steady-state axial temperature profiles at varying heating input power values and various cooling methods for sodium heat pipe coupled to a thermoelectric generator (Ref. 105).

A high degree of vacuum was concluded to be necessary to achieve excellent heat transfer performance of the heat pipe. Also, the high capillary driving force, suction force, and reasonable filling ratio of working fluid were considered critical factors in the heat pipe's adaptive heat transfer capacity. The heat pipe diameter also influenced the heat pipe's adaptive heat transfer capacity capability.

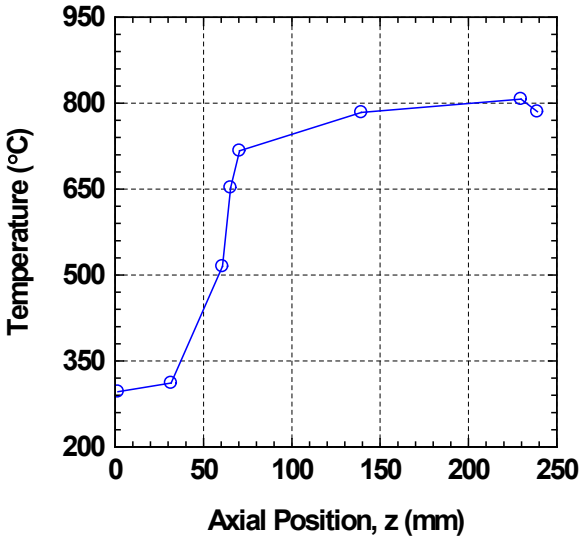


Figure 23. Axial temperature profile of big diameter HTHP in anti-gravity conditions (Ref. 108).

Sun investigated the thermal behavior of a horizontal high-temperature potassium heat pipe under transitory inclination, vertical heave, and periodic swing for heat pipe cooled reactors (Ref. 109). The stainless steel 316 heat pipe with a wick structure consisting of stainless steel 800-mesh screens was investigated. The evaporator section was heated using an electric heater, and the condenser section was cooled by natural air convection. The steady-state axial wall temperature profiles with varying lengths of the condenser (250-mm and 370-mm) were reported for Q_{in} = 774 W, as shown in Fig. 24. The transitory inclination was observed to cause an apparent temperature variation in both the evaporator and condenser sections. The vertical heave (motion direction perpendicular to heat pipe) had negligible effects. The periodic swing caused periodic temperature fluctuations. The dryout phenomenon of the heat pipe was achieved more efficiently when the condenser section was longer.

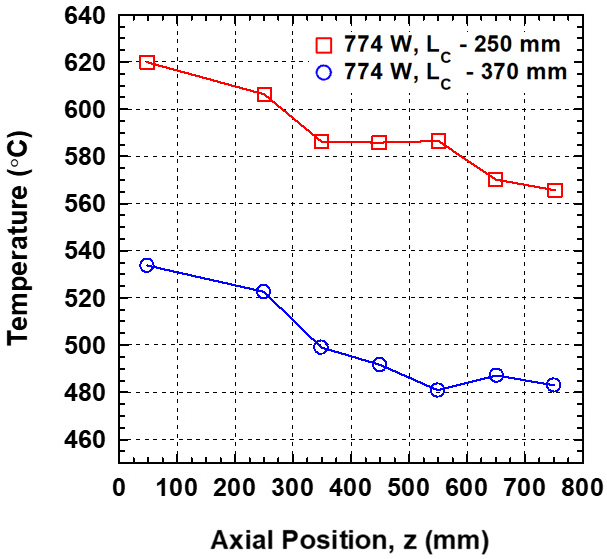


Figure 24. Axial wall temperature profile with varying condenser lengths and constant power load for potassium heat pipe (Ref. 109).

III.C. Other nuclear-related technologies

The only reported experimental data provided for nuclear technologies aside from reactor systems are spent fuel pools, though there is overlap in reactor cooling with its usage in spent fuel storage pools. At the end of this section, a table (Table III) lists references related to the application of heat pipes in other nuclear-related technology presented herein.

Even though the amount of decay heat of SFP is much less than the heat released in the actual nuclear reactor core, it is imperative to remove this heat. If not removed, the accumulation of heat can damage the integrity of the spent fuel cladding. Additionally, an excess of heat can dry out the spent fuel pool, the spent fuel to melt, and cause hydrogen gas accumulation, which increases the probability of explosion (Ref. 166). Thermal energy from spent nuclear fuel worldwide is ~750 MWth but is usually removed by chilling the water or air in dry storage casks (Ref. 124). Spent fuel of PWRs is placed in intermediate storage pools for about 3-5 years (Ref. 167) and then air-cooled in dry storage casks for ~40-60 years (Ref. 124). Studies have proved that heat can be harnessed by placing heat pipes at various stages of the spent fuel's decay heat removal process, which is utilized to generate electricity.

Table II: Experimental studies containing data sets pertaining to microreactor and terrestrial reactor technologies.

Reference	Working Fluid	Operating Temperature	Wall Material	Wick Design	Dimensions	Heater Power (W)
Seo, 2016 (Ref. 118)	Distilled water	100-175°C	Stainless steel	Wickless	D_o = 19.05 mm D_i = 16.57 mm L_t = 3000 mm L_e = 1000 mm L_c = 1000 mm	600-2100
Kim, 2017 (Ref. 123)	Water	65-240°C	Stainless steel 316L	Wickless with annular B_4C fill section	D_o = 25.4 mm D_i = 22 mm L_t = 285 mm L_e = 215 mm L_c = 500 mm	200-5800
Kim, 2017 (Ref. 62)	Water/FC-72	40-50°C	Stainless steel	Various wick structures	D_o = 12 mm D_i = 10 mm L_t = 650 mm L_e = 200 mm L_c = 300 mm	30-130
Panda, 2017 (Ref. 150)	Sodium	750°C	Stainless steel 304	Four layers 100-mesh stainless steel wire screen	L_{side} = 25.07 mm D_i = 21.87 mm L_t = 400 mm L_e = 150 mm L_c = 50 mm	3815.15
Tang, 2020 (Ref. 147)	Potassium	577-585°C	-	300-mesh	D_o = 28 mm D_i = 22 mm L_t = 800 mm L_e = 200 mm L_c = 250 mm	400

Reference	Working Fluid	Operating Temperature	Wall Material	Wick Design	Dimensions	Heater Power (W)
Teng, 2020 (Ref. 103)	Sodium	850°C	Stainless steel 310	Double-layer of SS 20-mesh wire	D_o = 38 mm D_i = 30 mm L_t = 800 mm L_e = 450 mm L_c = 200 mm	2500
Wang, 2020 (Ref. 105)	Sodium	530-625°C	Stainless steel 316L	300-mesh stainless steel screen	D_o = 28 mm D_i = 22 mm L_t = 800 mm L_e = 200 mm L_c = 250 mm	320-1550
Ji, 2021 (Ref. 108)	Sodium	800°C	Stainless steel 310S	Two outer layers of 400 mesh and two middle layers of 200-mesh	D_o = 100 mm D_i = 52 mm L_t = 240 mm L_e = 60 mm L_c = 180 mm	999
Wang, 2021 (Ref. 106)	Potassium	425-625°C	Stainless steel 316	300-mesh stainless steel 316 screen	D_o = 30 mm D_i = 23 mm L_t = 800 mm L_e = 230 mm L_c = 370 mm	370-710
Sun, 2022 (Ref. 109)	Potassium	530-620°C	Stainless steel 316	800-mesh stainless steel 316 screen	D_o = 30 mm D_i = 21 mm L_t = 800 mm L_e = 250 mm L_c = 250, 370 mm	774

Subscripts: t: total, e: evaporator, o: outer, i: inner, and c: condenser

Kim experimentally investigated vertical and horizontal water heat pipes for passive fuel cooling (Ref. 126). The stainless steel 316L heat pipe was 150-mm long with a 6.65-mm OD and 1-mm wall thickness. The wick structure was a double-layer of stainless steel 100-mesh screen wire with an approximate porosity of 0.62. The non-condensable gas was removed using a vacuum pump, and the fill ratio of distilled water was 80%. A direct current between copper electrodes heated the 60-mm evaporator region. A water jacket cooled the 60-mm condenser section. The adiabatic section was insulated using glass wool. The steady-state axial wall temperature profiles for vertical and horizontal conditions were provided at various heating input power values, as shown in Fig. 25. The heat pipe's overall heat resistance and heat transfer coefficient were also provided according to heat input. The vertical heat pipes showed better heat transfer performance than horizontal heat pipes due to higher heat transfer coefficients because of gravitational force.

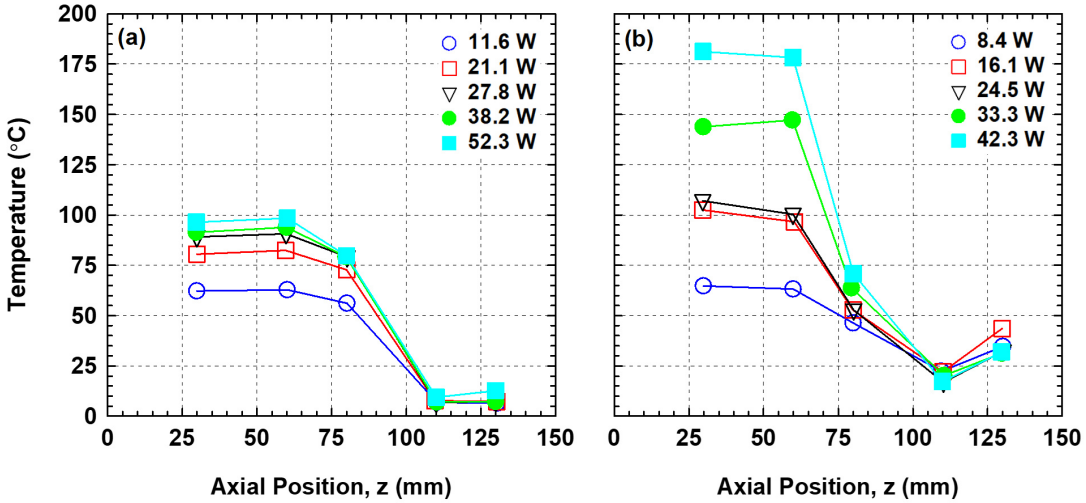


Figure 25. Steady-state temperature profiles in the (a) horizontal and (b) vertical positions at various heat input values for water heat pipes (Ref. 126).

Rosidi investigated the effects of graphenano-fluid on the heat pipe thermal performance for passive heat removal in nuclear spent fuel storage pools (Ref. 85). Graphenenano-fluid with a weight concentration of 1% as the working fluid was added into the heat pipe. The working fluid was added to a vacuous heat pipe with a filling ratio of 80%. The wickless copper heat pipe has an evaporator section heated using eight heater belts. The condenser section was cooled using a water jacket with a coolant flow rate of 8 L/min, circulating water from the thermostatic bath to the condenser. The steady-state axial wall temperature profile for various axial heat input values of 1000-2500 W was reported in Fig. 26. Additional information includes a transient temperature profile with a filling ratio of 80% and a cooling flow rate of 8 L/min, and the thermal resistance of the wickless-heat pipe as a function of the heat load. The graphenano-fluid was shown to enhance the heat pipe thermal performance.

Kusuma investigated the thermal performance of a vertical two-phase closed thermosyphon as a passive cooling system for a nuclear reactor spent fuel storage pool (Ref. 122). The wickless copper heat pipe was heated using a Ni-Chrome wire resistance heater veiled with ceramic. The condenser section was cooled using a water jacket connected to a circulating thermostatic bath. The pressure variations inside the heat pipe were investigated for -62, -64, and -66 cmHg, as shown in Fig. 27. The steady-state axial wall temperature profile for the two-phase closed thermosyphon was provided with a filling ratio of 60%, a heat load of 187 W, and a cooling

water mass flow rate of 1 L/min and 30°C constant temperature, as shown in Fig. 27. Additional information includes the transient temperature profile for a filling ratio of 60% and a heat loss of 187 W and the thermal resistance of the thermosyphon with various filling ratios (45-70%) as a function of heat load. The best performance was achieved with the lowest initial pressure at a filling ratio of 60%.

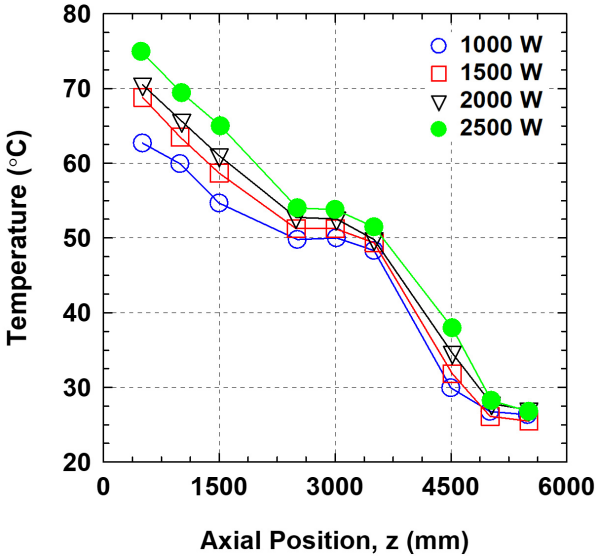


Figure 26. Steady-state axial wall temperature profiles for graphenenano-fluid enhanced thermosyphons at various evaporator heat loads (Ref. 85).

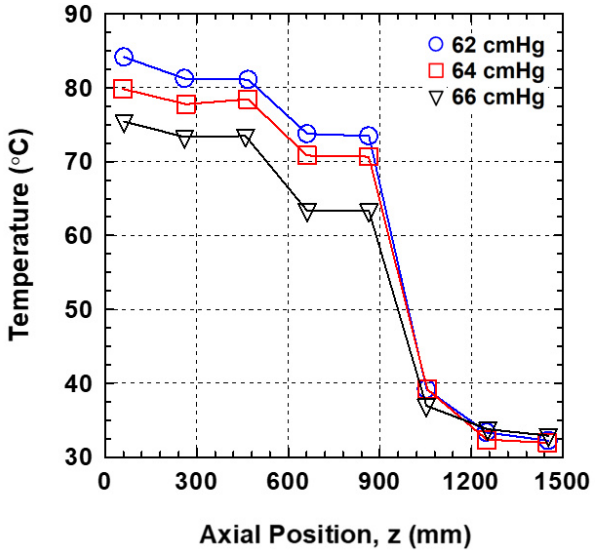


Figure 27. Axial wall temperature profile at various pressures for copper thermosyphon (Ref. 122).

Kusuma also investigated a vertical, straight, wickless-heat pipe's thermal performance to enhance thermal safety when a station blackout accident occurs (Ref. 86). The working fluid inside the tested heat pipe was demineralized water. The wickless copper heat pipe was heated using an electric band heater with a power source of 2800 W. The condenser section was cooled using a

water jacket connected to a circulating thermostatic bath. Various volumetric flow rates of the condenser coolant were varied at 4-8 L/min. Filling ratios of 40- 80% were tested. Various heat input power levels were reported (1000-2500 W), as shown in Fig. 28. Additional information regarding the thermal resistance as a function of heat load was provided for various initial pressures (2666, 15998, and 29330Pa) and transient temperature profiles as a function of time for various filling ratios and varied heat loads and coolant volumetric flow rates. The thermal resistance as a function of the heat load was also provided for various filling ratios with a volumetric coolant flow rate of 8 L/min. The experimental results showed that the best thermal performance was obtained at a thermal resistance of about 0.016 °C/W, a filling ratio of 80%, a lower initial pressure, a higher coolant volumetric flow rate, and a higher heat load. Also, experiments showed that the vertical, straight, wickless-heat pipe had higher thermal performance and can be proposed as a passive residual heat removal system of a nuclear spent fuel pool under station blackout.

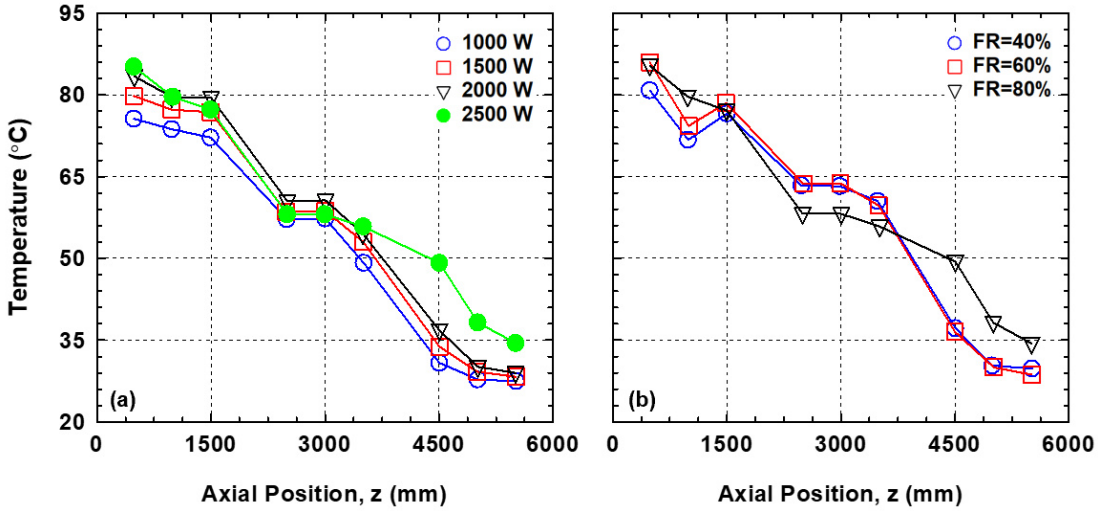


Figure 28. Steady-state axial wall temperature profiles of 80% filling ratios and heat load for a water thermosyphon (Ref. 86).

Kim investigated the usage of hybrid heat pipe systems in achieving electricity through a dry storage cask (UCAN) with refrigerant heat pipes and a Stirling engine (Ref. 124). The main components of the UCAN are hybrid heat pipes with 21 evaporation sections with neutron-absorbing materials and a single condenser section. Three dry storage casks were tested: UCAN-B, UCAN-HP, and UCAN-HS. The UCAN-B was the traditional dry storage cask with regular heat pipes. The UCAN-HP utilized the proposed hybrid heat pipes at the center of each fuel assembly and the top of the dry storage cask. The UCAN-HS utilized the proposed hybrid heat pipes with the Stirling engine. The wickless copper hybrid heat pipe contained an annular B_4C in the evaporator section. The evaporator section of the UCAN was heated using twenty-one cartridge heaters installed in the canister wall (35-65 W). For the UCAN-B and UCAN-HP, the condenser section was cooled using external air. For the UCAN-HS, the condenser section was cooled using the Stirling engine. Steady-state axial wall temperature profile results for different UCANs are shown in Fig. 29. The electricity generation of the Stirling engine was 2.3% to 3.2% of the total heat load. Results also proved that the heat generated from spent nuclear fuel (SNF) is a significant electrical source and that the technology will become more commonly used as the number of SNF increases due to the continued operation of nuclear power plants.

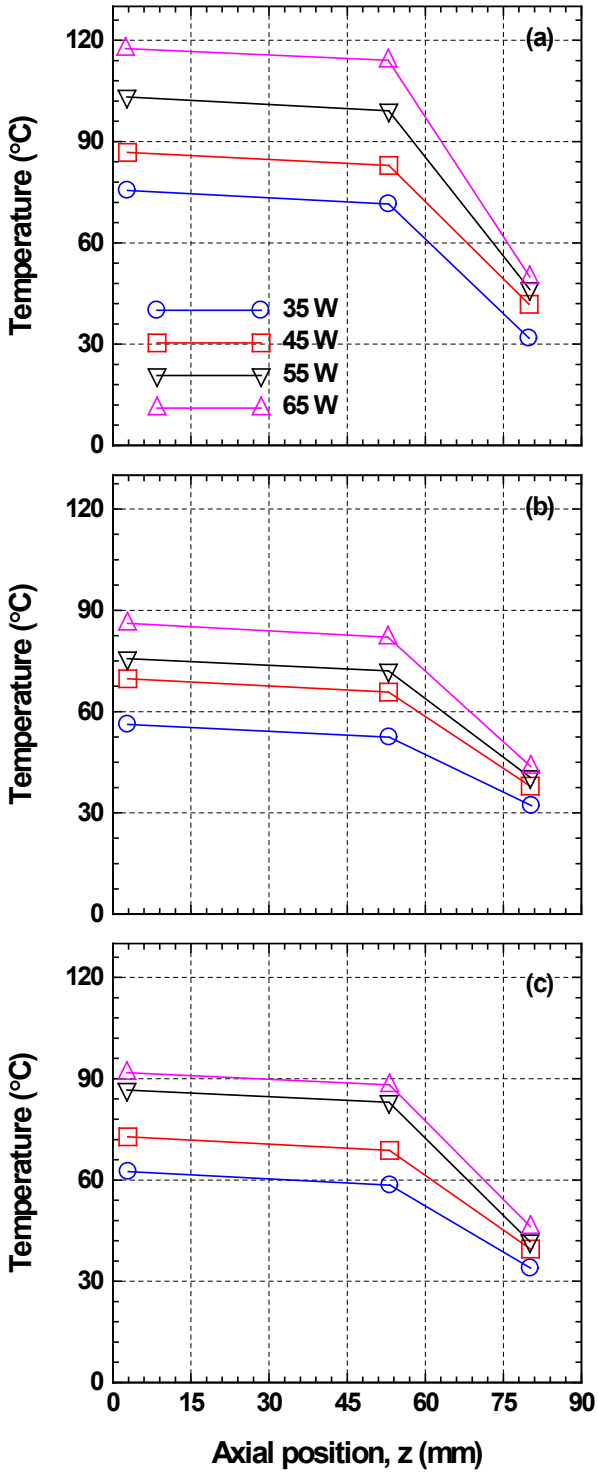


Figure 29. Steady-state axial wall temperature distribution of radial heat transfer path for various UCAN concepts (a) UCAN-B, (b) UCAN-HP, (c) UCAN-HS (Ref. 124).

Table III: Experimental studies containing data sets of other nuclear-related technologies.

Reference	Working Fluid	Operating Temperature	Wall Material	Wick Design	Dimensions	Heater Power (W)
Kim, 2014 (Ref. 126)	Distilled water	60-95°C	Stainless steel 316L	Double-layer of 100-mesh stainless steel screen wire	D_o = 6.65 mm D_i = 4.65 mm L_t = 150 mm L_e = 60 mm L_c = 60 mm	10-50
Rosidi, 2017 (Ref. 85)	H_2O-1%Graphene nano-fluid	62.5-75°C	Copper	Wickless	D_o = 103.1 mm D_i = 101.6 mm L_t = 6000 mm L_e = 1500 mm L_c = 2500 mm	1000-2500
Kusuma, 2017 (Ref. 122)	Demineralized water	75-85°C	Copper	Wickless	D_o = 26.4 mm D_i = 25.4 mm L_t = 1500 mm L_e = 500 mm L_c = 500 mm	187
Kusuma, 2018 (Ref. 86)	Demineralized water	80-85°C	Copper	Wickless	D_o = 107.6 mm D_i = 106.1 mm L_t = 6000 mm L_e = 2000 mm L_c = 2000 mm	2500
Kim, 2018 (Ref. 124)	Refrigerant gas	55-85°C	Copper	Wickless	$D_{o,e}$ = 6 mm $D_{o,c}$ = 12 mm L_t = 460 mm L_e = 400 mm L_c = 60 mm	35-65

Subscripts: t: total, e:evaporator, o: outer, i: inner, and c:condenser

IV. CONCLUSION

The adaptability of heat pipes has allowed for its comprehensive integration into multitudes of various nuclear technologies over the years, with applications still being discovered today. A review of several experimental heat pipe studies conducted for nuclear-related technological applications was presented to provide readers with potential opportunities for research topics regarding heat pipe technology in nuclear-related fields and equip researchers with potential data sets for validating heat pipe codes. Information regarding the presented heat pipes characteristics was presented in wick structure, wick type, and working fluid. Experimental heat pipe studies that detailed the heating and cooling methods were presented to provide researchers with an extensive list of heating and cooling options in their test facility. Heat pipe experimental data sets that contained pertinent information that may be beneficial to researchers were provided for thermal control systems, space nuclear radiators, direct electricity production through thermionic converters, emergency core cooling systems, direct cooling of the reactor core, hybrid heat pipe-control rods, and spent fuel pool waste's heat recovery systems. Heat pipes have shown to have a tremendously positive impact on nuclear technologies and will continue to become a more prevalent technology as more nuclear reactor concepts embrace this robust technology.

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REFERENCES

1. D. A. REAY, "The Perkins Tube – A Noteworthy Contribution to Heat Exchanger Technology," *Heat Recovery Sys.*, 2, 2, 173-187 (1982); https://doi.org/10.1016/0198-7593(82)90045-5.
2. R. S. GAUGLER, "Heat Transfer Device," *United States Patent Office,* (1944); https://lens.org/039-405-146-456-723.
3. L. M. TREFETHEN, "On the Surface Tension Pumping of Liquids, or, A Possible Role of the Candle Wick in Space Exploration," *General Electric Tech. Inform. Ser.*, No. 61SD114, (1962).
4. T. WYATT, "Satellite Temperature Stabilization System," *United States Patent Office*, (1964); https://patents.google.com/patent/US3152774A/en.
5. G. M. GROVER, T. P. COTTER, G. F. ERICKSON, "Structures of very high thermal conductance," *J. Applied Physics*, 35, 6 (1964); https://doi.org/10.1063/1.1713792.
6. G. M. GROVER, "Evaporation-Condensation Heat Transfer Device," *United States Patent Office*, (1966); https://www.lens.org/lens/patent/137-356-119-991-880.
7. G. M. GROVER, C. A. BUSSE, J. BOHDANKSY, "Nuclear Reactor with Thermionic Converter," *United States Patent Office,* (1967); https://lens.org/070-424-404-776-402.
8. J. E. DEVERALL, E. W. SALMI, R. J. KNAPP, "Orbital Heat Pipe Experiment," *Los Alamos National Lab.*, (1967); https://doi.org/10.2172/4349401.
9. P. FIEBELMANN, V. BESOZZO, "Nuclear Reactor," *United States Patent Office*, (1968); https://lens.org/114-677-671-705-043.
10. C. A. HEATH, E. LANTZ, "Nuclear Thermionic Space Power System Concept Employing Heat Pipes," *National Aeronautics and Space Administration*, (1968); https://www.osti.gov/biblio/4505844.
11. V. E. HAMPEL, R. P. KOOPMAN, "Reactivity Self-Control on Power and Temperature in Reactors Cooled by Heatpipes," *Trans. American Nucl. Soc.*, 11, 433-4 (1968); https://www.osti.gov/biblio/4813461.
12. G. NEIDERAUER, E. LANTZ, "Split-Core Heat Pipe Reactor Concept for an Out-of-Core Thermionic Power System," *IEEE Conf.*, 257-261 (1970); https://www.osti.gov/biblio/4717845.
13. G. NIEDERAUER, E. LANTZ, R. BREITWEISER, "Split-Core Heat-Pipe Reactors for Out-of-Pile Thermionic Power Systems," *Trans. American Nucl. Soc.*, (1971); https://ntrs.nasa.gov/citations/19720030759.
14. D. R. KOENIG, W. A. RANKEN, E. W. SALMI, "Heat Pipe Reactors for Space Power Applications," *J. Energy*, 1, 4 (1977); https://doi.org/10.2514/3.62334.
15. V. C. TRUSCELLO, "SP-100, The U.S. Space Nuclear Reactor Power Program. Technical information report," *National Aeronautics and Space Administration Jet Propulsion Lab*, (1983); https://doi.org/10.2172/10184691.
16. V. F. DEAN, M. S. EL-GENK, D. L. Y. LOUIE, D. M. WOODALL, "Heat Pipe Space Nuclear Reactor Design Assessment. Volume 1. Design status of the SP-100 Heat-Pipe Space Nuclear

Reactor system. Final report, 5 May 1982-27 March 1985," *Air Force Weapons Laboratory*, (1985); https://www.osti.gov/biblio/5995869.
17. M. P. MORIARTY, W. R. DETERMAN, "SP-100 Advanced Radiator Designs for Thermoelectric and Stirling Applications," *Rockwell International/Rocketdyne Division*, (1989); https://doi.org/10.1063/1.40128.
18. R. D. ROVANG, M. E. HUNT, R. B. DIRLING, R. A. HOLZL, "SP-100 High-Temperature Advanced Radiator Development," *AIP Conference Proceedings*, 217, 702 (1991); https://doi.org/10.1063/1.40128.
19. A. J. JUHASZ, R. D. ROVANG, "Development of lightweight prototype carbon-carbon heat pipe with integral fins and metal foil liner," *AIP Conference Proceedings,* 324, 135 (1995); https://doi.org/10.1063/1.47214.
20. R. B. HARTY, L. S. MASON, "100-kWe Lunar/Mars surface power utilizing the SP-100 reactor with dynamic conversion," *AIP Conference Proceedings*, 271, 1065 (1993); https://doi.org/10.1063/1.43087.
21. A. T. MARRIOTT, T. FUJITA, "Evolution of SP-100 System Designs," *AIP Conference Proceedings*, 301, 157 (1994); https://doi.org/10.1063/1.2950180.
22. A. T. JOSLOFF, N. F. SHEPARD, T. S. CHAN, F. C. GREENWOOD, N. A. DEANE, J. D. STEPHEN, R. E. MURATA, "SP-100 Generic Flight System Design and Early Flight Options," *AIP Conference Proceedings*, 301, 533 (1994); https://doi.org/10.1063/1.2950231.
23. G. E. POWELL, F. J. REDD, "Thermion: Verification of a Thermionic Heat Pipe in Microgravity," *Utah State University*, (1991); https://www.osti.gov/biblio/7239615.
24. W. A. RANKEN, J. A. TURNER, "Design Studies of the Moderated Heat Pipe Thermionic Reactor (MOHTR) Concept," *26th Intersociety Energy Conversion Engineering Conference*, (1991); https://digital.library.unt.edu/ark:/67531/metadc1092499/.
25. M. A. MERRIGAN, V. L. TRUJILLO, "Moderated heat pipe thermionic reactor (MOHTR) module development and test," *AIP Conference Proceedings*, 246, 1038 (1992); https://doi.org/10.1063/1.41781.
26. L. L. BEGG, T. J. WUCHTE, W. D. OTTING, "Star-C thermionic space nuclear power system," *AIP Conference Proceedings*, 246, 114 (1992); https://doi.org/10.1063/1.41774.
27. W. R. DETERMAN, G. HAGELSTON, "Thermionic in-core heat pipe design and performance," *AIP Conference Proceedings*, 246, 1046 (1992); https://doi.org/10.1063/1.41913.
28. W. A. RANKEN, M. G. HOUTS, "Heat pipe cooled reactors for multi-kilowatt space power supplies," *Los Alamos National Lab.*, (1995); https://www.osti.gov/biblio/10115990.
29. M. G. HOUTS, D. I. POSTON, W. A. RANKEN, "HPS: A Space Fission Power System Suitable for Near-Term, Low-Cost Lunar and Planetary Bases," *Fifth International Conference and Exposition of Engineering, Construction, and Operations in Space*, (1996).
30. D. I. POSTON, M. G. HOUTS, "Nuclear and Thermal Analysis of the Heatpipe Power and Bimodal Systems," *AIP Conference Proceedings*, 361, 1129 (1996); https://doi.org/10.1063/1.50052.
31. D. I. POSTON, "The Heatpipe-Operated Mars Exploration Reactor (HOMER)," Los Alamos National Lab., (2000); https://www.osti.gov/servlets/purl/766062.
32. R. S. REID, J. T. SENA, A. L. MARTINEZ, "Sodium heat pipe module test for the SAFE-30 reactor prototype," *AIP Conference Proceedings*, 552, 869 (2001); https://doi.org/10.1063/1.1358021.

33. D. I. POSTON, R. J. KAPERNICK, R. M. GUFFEE, "Design and analysis of the SAFE-400 space fission reactor," *AIP Conference Proceedings*, 608, 578 (2002); https://doi.org/10.1063/1.1449775.
34. P. J. RING, E. D. SAYRE, J. T. SENA, "SAFE-100 Module Fabrication and Test," *AIP Conference Proceedings*, 654, 96 (2003); https://doi.org/10.1063/1.1541282.
35. B. E. STEEVE, "Safe Affordable Fission Engine- (SAFE-) 100a Heat Exchanger Thermal and Structural Analysis," *National Aeronautics and Space Administration*, (2005); https://ntrs.nasa.gov/citations/20050166900.
36. M. S. EL-GENK, J.-M. TOURNIER, "SAIRS-Scalable Amtec Integrated Reactor Space Power System," *Prog. Nucl. Energy*, 45, 1, 25-69 (2004); https://doi.org/10.1016/j.pnucene.2004.08.002.
37. P. C. SCHMITZ, J. G. SCHREIBER, L. B. PENSWICK, "Feasibility Study of a Nuclear-Stirling Power Plant for the Jupiter Icy Moons Orbiter," *AIP Conference Proceedings*, 746, 738 (2005); https://doi.org/10.1063/1.1867193.
38. J.-M. TOURNIER, M. S. EL-GENK, "Liquid Metal Heat Pipes Radiator for Space Nuclear Reactor Power Systems," *American Physics Society – Power Component/Sub-System Integration*, (2005); https://doi.org/10.2514/6.2005-5701.
39. M. S. EL-GENK, J.-M. TOURNIER, "High Temperature Water Heat Pipes Radiator for a Brayton Space Reactor Power System," *AIP Conference Proceedings*, 813, 716 (2006); https://doi.org/10.1063/1.2169253.
40. J. C. KING, M. S. EL-GENK, "Submersion-Subcritical Safe Space (S^4) reactor," *Nucl. Eng. Des.*, 236, 17, 1759-1777 (2006); https://doi.org/10.1016/j.nucengdes.2005.12.010.
41. B. M. GALLO, M. S. EL-GENK, "Brayton rotating units for space reactor power systems," *Energy Conversion and Management*, 50, 9, 2210-2232 (2009); https://doi.org/10.1016/j.enconman.2009.04.035.
42. J. R. HARTENSTINE, W. G. ANDERSON, R. BONNER, "Titanium Loop Heat Pipes for Space Nuclear Power Systems," *AIP Conference Proceedings*, 969, 44 (2008); https://doi.org/10.1063/1.2845001.
43. E. GREENSPAN, "A Solid-Core Heat-Pipe Nuclear Battery Type Reactor," *University of California, Berkeley*, (2008); https://www.osti.gov/servlets/purl/940911.
44. J.D. BESS, "A Basic LEGO Reactor Design for the Provision of Lunar Surface Power," Idaho National Lab., (2008); https://www.osti.gov/servlets/purl/935462-WNhlWU/.
45. M. EADES, J. FLANDERS, T. BLUE, X. SUN, "Space Molten Salt Reactor Design Considerations and Research Needs," *Nucl. Emerg. Tech. Space*, 3070 (2012).
46. M. E. BETTENCOURT, "Fluidic and Neutronic Coupled Modeling of the Space Molten Salt Reactor Concept," *Ohio State University*, (2013).
47. Y. YUAN, J. SHAN, B. ZHANG, J. GOU, B. ZHANG, T. LU, L. GE, Z. YANG, "Study on startup characteristics of heat pipe cooled and AMTEC conversion space reactor system," *Prog. Nucl. Energy*, 86, 18-30 (2016); https://doi.org/10.1016/j.pnucene.2015.10.002.
48. X. WANG, Q. ZHANG, K. ZHUANG, X. HE, M. SEIDL, R. JUAN-MACIAN, "Neutron physics of the liquid-fuel heat-pipe reactor concept with molten salt fuel – Static calculations," *Int. J. Energy Res.*, 43, 14, 7852-7865 (2019); https://doi.org/10.1002/er.4642.
49. M. A. GIBSON, L. S. MASON, C. L. BOWMAN, D. I. POSTON, P. R. MCCLURE, J. CREASY, C. ROBINSON, "Development of NASA's Small Fission Power System for Science and Human Exploration," *National Aeronautics and Space Administration*, (2015); https://ntrs.nasa.gov/citations/20140017750.

50. D. BEARD, C. TARAU, W. G. ANDERSON, "Sodium Heat Pipes for Spacecraft Fission Power Generation," *15th International Energy Conversion Engineering Conference*, (2017); https://doi.org/10.2514/6.2017-5087.
51. M. A. GIBSON, S. R. OLESON, D. I. POSTON, P. MCCLURE, "NASA's Kilopower Reactor Development and the Path to Higher Power Missions," *National Aeronautics and Space Administration*, (2017); https://ntrs.nasa.gov/citations/20170002010.
52. D. I. POSTON, M. GIBSON, P. MCCLURE, "Kilopower Reactors for Potential Space Exploration Missions," American Nuclear Society Topical Meeting, (2019).
53. D. I. POSTON, M. A. GIBSON, R. G. SANCHEZ, P. R. MCCLURE, "Results of the KRUSTY Nuclear System Test," *Nucl. Tech.*, 206, 1, S89-S117 (2020); https://doi.org/10.1080/00295450.2020.1730673.
54. P. R. HUEBOTTER, G. A. MCLENNAN, "Fast Reactor Power Plant Design Having Heat Pipe Heat Exchanger," *United States Patent Office*, (1984); https://www.lens.org/lens/patent/088-538-890-111-13X.
55. V. E. HAMPEL, "Underground nuclear power station using self-regulating heat-pipe controlled reactors," *United States Patent Office*, (1989); https://www.osti.gov/biblio/5197028.
56. K. OHASHI, H. HAYAKAWA, M. YAMADA, "Preliminary study on the application of the heat pipe to the passive decay heat removal system of the modular HTR," *Prog. Nucl. Energy*, 32, 3-4, 587-594 (1998). https://doi.org/10.1016/S0149-1970(97)00047-4.
57. S.-H. KIM, K. K. KIM, J. W. YEO, M. H. CHANG, S. Q. ZEE, "Design Verification Program of SMART," GENES4/ANP2003, 1047 (2003).
58. M. MOCHIZUKI, T. NGUYEN, R. SINGH, T. NGUYEN, S. SUGIHARA, K. MASHIKO, Y. SAITO, V. WUTTIJUMNONG, "Nuclear Reactor Must Need Heat Pipe for Cooling," *Front. Heat Pipes*, 2, 033001 (2011).
59. C. WANG, D. ZHANG, S. QIU, W. TIAN, Y. WU, G. SU, "Study on the characteristics of the sodium heat pipe in passive residual heat removal system of molten salt reactor," *Nucl. Eng. Des.*, 265, 691-700 (2013); https://doi.org/10.1016/j.nucengdes.2013.09.023.
60. C. WANG, Z. GUO, D. ZHANG, S. QIU, W. TIAN, Y. WU, G. SU, "Transient behavior of the sodium-potassium alloy heat pipe in passive residual heat removal system of molten salt reactor," *Prog. Nucl. Energy*, 68, 142-152 (2013); https://doi.org/10.1016/j.pnucene.2013.07.001.
61. Y. S. JEONG, K. M. KIM, I. G. KIM, I. C. BANG, "Feasibility study of hybrid heat pipe with control rod as passive in-core cooling system for advanced nuclear power plant," *NURETH-16 Conference*, (2015).
62. K. M. KIM, I. C. BANG, "Thermal-hydraulic phenomena inside hybrid heat pipe-control rod for passive heat removal," *Int. J. Heat Mass Transf.*, 119, 472-483 (2018); https://doi.org/10.1016/j.ijheatmasstransfer.2017.11.138.
63. K. M. KIM, I. C. BANG, "Heat transfer characteristics and operation limit of pressurized hybrid heat pipe for small modular reactors," *Appl. Therm. Eng.*, 112, 560-571 (2017); https://doi.org/10.1016/j.applthermaleng.2016.10.077.
64. P. R. MCCLURE, D. I. POSTON, V. R. DASARI, R. S. REID, "Design of Megawatt Power Level Heat Pipe Reactors," *Los Alamos National Lab.*, (2015); https://doi.org/10.2172/1226133.
65. I. I. SVIRIDENKO, D. V. SHEVELYOV, O. V. POLYAKOV, V. A. TIMOFEEV, N. N. SVIRIDENKO, "Passive Residual Heat Removal System for WWER with the Thermosiphon

Heat Exchange Equipment," *Int. J. Energy Clean Env.*, 16, 1-4, 209-223 (2015); DOI: 10.1615/InterJEnerCleanEnv.2015015683.
66. J. W. STERBENTZ, J. E. WERNER, M. G. MCKELLAR, A. J. HUMMEL, J. C. KENNEDY, R. N. WRIGHT, J. M. BIERSDORF, "Special Purpose Nuclear Reactor (5 MW) for Reliable Power at Remote Sites Assessment Report," *Idaho National Lab.*, (2017); https://doi.org/10.2172/1410224.
67. J. W. STERBENTZ, J. E. WERNER, A. J. HUMMEL, J. C. KENNEDY, R. C. O'BRIEN, A. M. DION, R. N. WRIGHT, K. P. ANANTH, "Preliminary Assessment of Two Alternative Core Design Concepts for the Special Purpose Reactor," *Idaho National Lab.*, (2018); https://doi.org/10.2172/1413987.
68. A. LEVINSKY, J. J. V. WYK, Y. ARAFAT, M. C. SMITH, "Westinghouse eVinci^{TM} Reactor for Off-Grid Markets," *Trans. American Nucl. Soc.*, 119, 931-934 (2018).
69. J. C. KENNEDY, P. SABHARWALL, K. FRICK, M. MCKELLAR, S. BRAGG-SITTON, "Special Purpose Application Reactors: System Integration Decision Support," *Idaho National Lab.*, (2019); https://doi.org/10.2172/1475413.
70. P. SABHARWALL, C.-S. LIN, J. HANSEL, V. LABOURE, D. ANDRS, W. HOFFMAN, S. NOVASCONE, A. SLAUGHTER, R. MARTINEAU, "Application of Integrated Modeling and Simulation Capabilities for Full Scale Multiphysics Simulation of Microreactor Concepts," *Idaho National Lab.*, (2019); https://inldigitallibrary.inl.gov/sites/sti/sti/Sort_19803.pdf.
71. R. HERNANDEZ, M. TODOSOW, "Micro heat pipe nuclear reactor concepts: Analysis of fuel cycle performance and environmental impacts," *Annals Nucl. Eng.*, 126, 419-426 (2019); https://doi.org/10.1016/j.anucene.2018.11.050.
72. C.-Y. XIE, H.-Z. TAO, W. LI, J.-J. CHENG, "Numerical simulation and experimental investigation of heat pipe heat exchanger applied in residual heat removal system," *Annals Nucl. Eng.*, 133, 568-579 (2019); https://doi.org/10.1016/j.anucene.2019.07.009.
73. B. CAI, S. CHEN, T. WU, Y. WANG, Z. ZHANG, C. YUAN, C. ZHANG, Y. GE, "Neutronic Design for Heat Pipe Reactor with Annular and Accident Tolerant Fuels," *Front. Energy Res.*, 9, 678262 (2021); https://doi.org/10.3389/fenrg.2021.678262.
74. D. Y. CUI, Y. DAI, X. Z. CAI, Y. FU, X. X. LI, Y. ZOU, J. G. CHEN, "Preconceptual nuclear design of a 50 kWth heat pipe cooled micro molten salt reactor (micro-MSR)," *Prog. Nucl. Energy*, 134, 103670 (2021); https://doi.org/10.1016/j.pnucene.2021.103670.
75. C. HU, H. WANG, B. WU, X. ZHANG, P. ZHANG, "Effect of heat pipe failure on performance of residual heat removal system with heat pipe for small lead-based reactor," *E3S Web Conf.*, 248, 01021 (2021); https://doi.org/10.1051/e3sconf/202124801021.
76. D. WANG, Y. GUO, "Preliminary Design of a Heat Pipe-Cooled Blanket for CFETR," *Energies*, 14, 21, 6879 (2021); https://doi.org/10.3390/en14216879.
77. G. A. EL-AWADI, S. ABDEL-SAMAD, M. ABDEL-BARY, J. RITMAN, "Improving the performance of the cryogenic heat pipe-target system for the COSY-TOF experiment," *Vacuum*, 83, 11, 1321-1325 (2009); https://doi.org/10.1016/j.vacuum.2009.04.039.
78. R. E. NYGREN, G. F. MATTHEWS, T. W. MORGAN, S. A. SILBURN, J. H. ROSENFELD, M. T. NORTH, A. TALLARIGO, V. N. STAVILA, "Post-test examination of a Li-Ta heat pipe exposed to H plasma in Magnum PSI," *Fusion Eng. Des.*, 146, Part B, 2603-2607 (2019); https://doi.org/10.1016/j.fusengdes.2019.04.053.

79. Y. BOUAICHAOUI, A. BELKAID, S. A. AMZERT, "Economic and Safety Aspects in Nuclear Seawater Desalination," *Procedia Eng.*, 33, 146-154 (2012); https://doi.org/10.1016/j.proeng.2012.01.1187.
80. H. JOUHARA, V. ANASTAOV, I. KHAMIS, "Potential of heat pipe technology in nuclear seawater desalination," *Desalination*, 249, 3, 1055-1061 (2009); https://doi.org/10.1016/j.desal.2009.05.019.
81. Z. XIONG, H. GU, M. WANG, Y. CHENG, "The thermal performance of a loop-type heat pipe for passively removing residual heat from spent fuel pool," *Nucl. Eng. Des.*, 280, 262-268 (2014); https://doi.org/10.1016/j.nucengdes.2014.09.022.
82. Z. XIONG, M. WANG, H. GU, C. YE, "Experimental study on heat pipe heat removal capacity for passive cooling of spent fuel pool," *Annals of Nucl. Energy*, 83, 258-263 (2015); https://doi.org/10.1016/j.anucene.2015.03.045.
83. Z. XIONG, C. YE, M. WANG, H. GU, "Experimental study on the sub-atmospheric loop heat pipe passive cooling system for spent fuel pool," *Prog. Nucl. Energy*, 79, 40-47 (2015); https://doi.org/10.1016/j.pnucene.2014.10.015.
84. Y. S. JEONG, I. C. BANG, "Hybrid heat pipe based passive cooling device for spent nuclear fuel dry storage cask," *Appl. Therm. Eng.*, 96, 277-285 (2016); https://doi.org/10.1016/j.applthermaleng.2015.11.086.
85. A. .ROSIDI, N. PUTRA, M. H. KUSUMA, "Effect of graphenenano-fluid on heat pipe thermal performance for passive heat removal in nuclear spent fuel storage pool," *IOP Conference Series: Earth and Environmental Science*, 105 (2017); https://iopscience.iop.org/article/10.1088/1755-1315/105/1/012030.
86. M. H. KUSUMA, N. PUTRA, A. R. ANTARIKSAWAN, R. A. KOESTOER, S. WIDODO, S. ISMARWANTI, B. T. VERLAMBANG, "Passive cooling system in a nuclear spent fuel pool using a vertical straight wickless-heat pipe," *Int. J. Therm. Sci.*, 126, 162-171 (2018); https://doi.org/10.1016/j.ijthermalsci.2017.12.033.
87. K. LAI, W. WANG, C. YI, Y. KUANG, C. YE, "The study of passive cooling system assisted with separate heat pipe for decay heat removal in spent fuel pool," *Annals Nucl. Eng.*, 111, 523-535 (2018); https://doi.org/10.1016/j.anucene.2017.08.062.
88. Y. KUANG, C. YI, W. WANG, "Heat transfer performance analysis of a large-scale separate heat pipe with a built-in tube," *Appl. Therm. Eng.*, 167, 114716 (2020); https://doi.org/10.1016/j.applthermaleng.2019.114716.
89. J. CHOI, C. LIM, H. KIM, "Fork-end heat pipe for passive air cooling of spent nuclear fuel pool," *Nuc. Eng. Des.*, 374, 111081 (2021); https://doi.org/10.1016/j.nucengdes.2021.111081.
90. M. S. EL-GENK, J.-M. P. TOURNIER, "Uses of Liquid-Metal and Water Heat Pipes in Space Reactor Power Systems," Front. Heat Pipes, 2, 013002 (2011).
91. A. FAGHRI, "Heat Pipes: Review, Opportunities and Challenges," Front. Heat Pipes, 5, 1 (2014).
92. A. FAGHRI, M. BUCHKO, Y. CAO, "A Study of High-Temperature Heat Pipes with Multiple Heat Sources and Sinks: Part I – Experimental Methodology and Frozen Startup Profiles," *ASME J. Heat Transf.*, 113, 4, 1003-1009 (1991); https://doi.org/10.1115/1.2911193.
93. J. H. JANG, "Startup Characteristics of a Potassium Heat Pipe from the Frozen State," *J. Thermophysics Heat Transf.*, 9, 1 (1995); https://doi.org/10.2514/3.636.
94. D. E. GLASS, M. A. MERRIGAN, J. T. SENA, "Start Up of a Nb-1%Zr Potassium Heat Pipe From the Frozen State," *National Aeronautics and Space Administration*, (1998).

95. Y. MA, H. YU, S. HUANG, Y. ZHANG, Y. LIU, C. WANG, R. ZHONG, X. CHAI, C. ZHU, X. WANG, "Effect of inclination angle on the startup of a frozen sodium heat pipe," *Appl. Therm. Eng.*, 201, 117625 (2022); https://doi.org/10.1016/j.applthermaleng.2021.117625.
96. H.-X. CHEN, Y.-X. GUO, D.-Z. YUAN, Y. JI, "Experimental study on frozen startup and heat transfer characteristics of a cesium heat pipe under horizontal state," *Int. J. Heat Mass Transf.*, 183, 122105 (2022); https://doi.org/10.1016/j.ijheatmasstransfer.2021.122105.
97. M. S. EL-GENK, L. HUANG, "An experimental investigation of the transient response of a water heat pipe," *Int. J. Heat Mass Transfer*, 36, 15, 3823-3830 (1993).
98. K. M. KIM, Y. S. JEONG, I. G. KIM, I. C. BANG, "Development of Passive In-Core Cooling System for Nuclear Safety Using Hybrid Heat Pipe," *Nucl. Technology*, 196, 3, 598-613 (2016); https://doi.org/10.13182/NT16-32.
99. I. G. KIM, K. M. KIM, Y. S. JEONG, I. C. BANG, "Performance of annular flow path heat pipe with a polymer insert controlling compactness for energy applications," *Int. J. Heat Mass Transf.*, 92, 929-939 (2016); https://doi.org/10.1016/j.ijheatmasstransfer.2015.09.037.
100. T. THARAYIL, L.G. ASIVATHAM, V. RAVINDRAN, S. WONGWISES, "Effect of filling ratio on the performance of a novel miniature loop heat pipe having different diameter transport lines," *Appl. Therm. Eng.*, 106, 588-600 (2016); https://doi.org/10.1016/j.applthermaleng.2016.05.125.
101. Y. WANG, J. CEN, F. JIANG, W. CAO, J. GUO, "LHP heat transfer performance: A comparison study about sintered copper powder wick and copper mesh wick," *Appl. Therm. Eng.*, 92, 104-110 (2016); https://doi.org/10.1016/j.applthermaleng.2015.08.109.
102. M. MOCHIZUKI, R. SINGH, T. NGUYEN, T. NGUYEN, "Heat pipe based passive emergency core cooling system for safe shutdown of nuclear power reactor," *Appl. Therm. Eng.*, 73, 1, 699-706 (2014); https://doi.org/10.1016/j.applthermaleng.2014.07.004.
103. W. TENG, X. WANG, Y. ZHU, "Experimental investigations on start-up and thermal performance of sodium heat pipe under swing conditions," *Int. J. Heat Mass Transf.*, 152, 119505 (2020); https://doi.org/10.1016/j.ijheatmasstransfer.2020.119505.
104. C. WANG, M. LIU, D. ZHANG, S. QIU, G.H. SU, W. TIAN, "Experimental study on transient performance of heat pipe-cooled passive residual heat removal system of a molten salt reactor," *Prog. Nucl. Energy*, 118, 103113 (2020); https://doi.org/10.1016/j.pnucene.2019.103113.
105. C. WANG, S. TANG, X. LIU, G.H. SU, X. TIAN, S. QIU, "Experimental study on heat pipe thermoelectric generator for industrial high temperature waste heat recovery," *Appl. Therm. Eng.*, 175, 115299 (2020); https://doi.org/10.1016/j.applthermaleng.2020.115299.
106. C. WANG, X. LIU, M. LIU, S. TANG, Z. TIAN, D. ZHANG, W. TIAN, S. QIU, G. SU, "Experimental study on heat transfer limit of high temperature potassium heat pipe for advanced reactors," *Annals Nucl. Eng.*, 151, 107935 (2021); https://doi.org/10.1016/j.anucene.2020.107935.
107. Z. TIAN, X. LIU, C. WANG, D. ZHANG, W. TIAN, S. QIU, G.H. SU, "Experimental investigation on the heat transfer performance of high-temperature potassium heat pipe for nuclear reactor," *Nucl. Eng. Des.*, 378, 111182 (2021); https://doi.org/10.1016/j.nucengdes.2021.111182.
108. Y. JI, D. YUAN, Z. CHE, J. ZHAO, D. TANG, Z. ZHANG, "Study on adaptive heat transfer performance of high temperature heat pipe," *Annals Nucl. Energy,* 163, 108536 (2021); https://doi.org/10.1016/j.anucene.2021.108536.

109. H. SUN, X. LIU, H. LIAO, C. WANG, J. ZHANG, W. TIAN, S. QIU, G. SU, "Experiment study on thermal behavior of a horizontal high-temperature heat pipe under motion conditions," *Annals Nucl. Energy*, 165, 108760 (2022); https://doi.org/10.1016/j.anucene.2021.108760.
110. W. LING, W. ZHOU, W. YU, R. LIU, K. S. HUI, "Thermal performance of loop heat pipes with smooth and rough porous copper fiber sintered sheets," *Energy Conversion and Management*, 153, 323-334 (2017); https://doi.org/10.1016/j.enconman.2017.10.009.
111. B. ZOHURI, "Heat Pipe Application in Fission Driven Nuclear Power Plant," *Int. J. Earth. Environ. Sci.*, 4, 166 (2019); https://doi.org/10.15344/2456-351X/2019/166.
112. K. NAKAMURA, K. ODAGIRI, H. NAGANO, "Study on a loop heat pipe for a long-distance heat transport under anti-gravity condition," *Appl. Therm. Eng.*, 107, 167-174 (2016); https://doi.org/10.1016/j.applthermaleng.2016.06.162.
113. J.-M. TOURNIER, M. S. EL-GENK, "Modeling of the startup of a horizontal lithium heat pipe from a frozen state," *Proc. NHTC'01 35th Nat. Heat Transf. Conf.*, (2001).
114. M. M SWARTZ, W. A. BYERS, J. LOJEK, R. BLUNT, "Westinghouse eVinci™ Heat Pipe Microreactor Technology Development," *Proc. 28th Conf. Nucl. Eng.*, (2021).
115. V. L. TRUJILLO, E. S. KEDDY, M. A. MERRIGAN, "High-Temperature, Deployable, Membrane Heat Pipe Radiator Element; Demonstration and Status," *Los Alamos National Lab.*, (1990); https://www.osti.gov/biblio/5383181.
116. H. MROUE, J. B. RAMOS, L. C. WROBEL, H. JOUHARA, "Experimental and numerical investigation of an air-to-water heat pipe-based heat exchanger," *Appl. Therm. Eng.*, 78, 339-350 (2015); https://doi.org/10.1016/j.applthermaleng.2015.01.005.
117. J. SEO, J.-Y. LEE, "Length effect on entrainment limitation of vertical wickless heat pipe," *Int. J. Heat Mass Transf.*, 101, 373-378 (2016); https://doi.org/10.1016/j.ijheatmasstransfer.2016.05.096.
118. J. SEO, I.-C. BANG, J.-Y. LEE, "Length effect on entrainment limit of large-L/D vertical heat pipe," *Int. J. Heat Mass Transf.*, 97, 751-759 (2016); https://doi.org/10.1016/j.ijheatmasstransfer.2016.02.065.
119. M. PETRUCCI, A. FAGHRI, "Multiple evaporator and condenser loop thermosyphon system for passive cooling of liquid-fuel molten salt nuclear reactors," *Nucl. Eng. Des.*, 370, 110936 (2020); https://doi.org/10.1016/j.nucengdes.2020.110936.
120. H. JOUHARA, S. ALMAHMOUD, D. BROUGH, V. GUICHET, B. DELPECH, A. CHAUHAN, L. AHMAD, N. SEREY, "Experimental and theoretical investigation of the performance of an air to water multi-pass heat pipe-based heat exchanger," *Energy*, 219, 119624 (2021); https://doi.org/10.1016/j.energy.2020.119624.
121. W. FU, X. LI, X. WU, Z. ZHANG, "Investigation of a long term passive cooling system using two-phase thermosyphon loops for the nuclear reactor spent fuel pool," *Annals Nucl. Eng.*, 85, 346-356 (2015); https://doi.org/10.1016/j.anucene.2015.05.026.
122. M. H. KUSUMA, N. PUTRA, A. R. ANTARIKSAWAN, SUSYADI, F. A. IMAWAN, "Investigation of the Thermal Performance of a Vertical Two-Phase Closed Thermosyphon as a Passive Cooling System for a Nuclear Reactor Spent Fuel Storage Pool," *Nucl. Eng. Tech.*, 49, 3, 476-483 (2017); https://doi.org/10.1016/j.net.2016.10.008.
123. I. G. KIM, K. M. KIM, Y. S. JEONG, I. C. BANG, "Flow visualization and heat transfer performance of annular thermosyphon heat pipe," *Appl. Therm. Eng.*, 125, 1456-1468 (2017); https://doi.org/10.1016/j.applthermaleng.2017.07.116.

124. I. G. KIM, I. C. BANG, "Spent nuclear fuel with a hybrid heat pipe for electricity generation and thermal management," *Energy Conversion and Management*, 173, 233-243 (2018); https://doi.org/10.1016/j.enconman.2018.07.084.
125. S. K. MUNIAPPAN, S. K. ARUMUGAM, "Experimental Investigations on an Axial Grooved Cryogenic Heat Pipe," Therm. Sci., 16, 1, 133-138 (2012); https://doi.org/10.2298/TSCI100805056S.
126. I. G. KIM, K. M. KIM, Y. S. JEONG, I. C. BANG, "Experimental Investigations of Vertical and Horizontal Heat Pipes," *Trans. Korean Nuclear Society Autumn Meeting*, (2014).
127. M. KUMAR, R. KANT, A. K. DAS, P. K. DAS, "Effect of Surface Tension Variation of the Working Fluid on the Performance of a Closed Loop Pulsating Heat Pipe," *Heat Transf. Eng.*, 40, 7, 509-523 (2019); https://doi.org/10.1080/01457632.2018.1436390.
128. J. E. DEVERALL, J. E. KEMME, L. W. FLORSCHUETZ, "Sonic Limitations and Startup Problems of Heat Pipes," *Los Alamos National Lab.*, (1970).
129. T. YAMAMOTO, K. NAGATA, M. KATSUTA, Y. IKEDA, "Experimental Study of Mercury Heat Pipe," Exp. Therm. Fluid Sci., 9, 1, 39-46 (1994); https://doi.org/10.1016/0894-1777(94)90006-X.
130. W. G. ANDERSON, J. H. ROSENFELD, D. ANGIRASA, Y. MI, "Evaluation of Heat Pipe Working Fluids in the Temperature Range 450 to 700 K," *AIP Conference Proceedings*, 699, 20 (2004); https://doi.org/10.1063/1.1649553.
131. X. ZHANG, K. HE, Q. YANG, C. XI, "Analysis on heat transfer and start-up performance of mercury heat pipe," *E3S Web Conf.*, 245, 03012 (2021); https://doi.org/10.1051/e3sconf/202124503012.
132. Z. Z. NEZAM, B. ZOHURI, "Heat Pipe as a Passive Cooling System Driving New Generation of Nuclear Power Plants," *Edelweiss Chem. Sci. J.*, 3, 1, 30-38 (2021); https://doi.org/10.33805/2641-7383.121.
133. C. HU, X. ZHANG, H. WAN, B. WU, P. ZHANG, "Analysis on start up and heat transfer performance of mercury heat pipe under alternating heating power," *E3S Web Conf,* 248, 01004 (2021); https://doi.org/10.1051/e3sconf/202124801004.
134. P. M. DUSSINGER, W. G. ANDERSON, "Design and Testing of Titanium/Cesium and Titanium/Potassium Heat Pipes," *Advanced Cooling Technologies*, (2005).
135. D. BUDEN, "Summary of Space Nuclear Reactor Power Systems, 1983-1992," *Idaho National Engineering Laboratory*, (1993); https://www.osti.gov/biblio/10151265.
136. K. A. WOLOSHUN, M. A. MERRIGAN, J. T. SENA, J. SECARY, "High temperature heat pipe experiments aboard the space shuttle," *AIP Conference Proceedings*, 271, 523 (1993); https://doi.org/10.1063/1.43064.
137. T. J. DICKINSON, "Performance Analysis of a Liquid Metal Heat Pipe Space Shuttle Experiment," *Air Force Institute of Technology*, (1996).
138. Y. Y. PARK, Y. S. JEONG, I. C. BANG, "Design Study of Heat Pipes for Nuclear Spaceship Applications," *Trans. Korea Nuclear Society Spring Meeting*, (2019).
139. H. ZOHURI, S. LAM, C. FORSBERG, "Heat-Pipe Heat Exchangers for Salt-Cooled Fission and Fusion Reactors to Avoid Salt Freezing and Control Tritium: A Review," *Nucl. Tech.*, 206, 11, 1642-1658 (2020); https://doi.org/10.1080/00295450.2019.1681222.
140. J.-M. TOURNIER, M. S. EL-GENK, "Startup of a horizontal lithium-molybdenum heat pipe from a frozen state," *Int. J. Heat Mass Transf.*, 46, 4, 671-685 (2003); https://doi.org/10.1016/S0017-9310(02)00324-1.

141. R. PONNAPPAN, "Studies on the Startup Transients and Performance of a Gas Loaded Sodium Heat Pipe," *Wright-Patterson Air Force Base*, (1989); https://www.osti.gov/biblio/7006928.
142. O. R. MIRELES, M. G. HOUTS, "Transient Response to Rapid Cooling of a Stainless-Steel Sodium Heat Pipe," *Proc. Nucl. Emerg. Tech. Space*, (2011); https://ntrs.nasa.gov/citations/20110008814.
143. X. JI, Y. WANG, J. XU, Y. HUANG, "Experimental study of heat transfer and start-up of loop heat pipe with multiscale porous wicks," *Appl. Therm. Eng.*, 117, 782-798 (2017); https://doi.org/10.1016/j.applthermaleng.2017.01.084.
144. R. LAUBSCHER, R. T. DOBSON, "Theoretical and experimental modeling of a heat pipe heat exchanger for high temperature nuclear reactor technology," *Appl. Therm. Eng.*, 61, 2, 259-267 (2013); https://doi.org/10.1016/j.applthermaleng.2013.06.063.
145. Y. XIONG, L. BO, M. QIANG, Y. WU, X. ZHANG, P. XU, C. MA, "A characteristic study on the start-up performance of molten-salt heat pipes: Experimental investigation," *Exp. Therm. Fluid Sci.*, 82, 433-438 (2017); https://doi.org/10.1016/j.expthermflusci.2016.11.031.
146. S.-W. KANG, H.-M. YEH, M.-C. TSAI, H.-H. WU, "Manufacture and Test of a High Temperature Heat Pipe," *J. Appl. Sci. Eng.*, 22, 3, 493-499 (2019).
147. S. TANG, C. WANG, D. ZHANG, W. TIAN, G. SU, S. QIU, "Thermoelectric performance study on a heat pipe thermoelectric generator for micro nuclear reactor application," *Int. J. Energy Res.*, 45, 8, 12301-12316 (2020); https://doi.org/10.1002/er.6450.
148. H.-M. LEE, M.-C. TSAI, H.-L. CHEN, H.-Y. LI, "Stainless Steel Heat Pipe Fabrication, Performance Testing and Modeling," *Energy Procedia*, 105, 4745-4750 (2017); https://doi.org/10.1016/j.egypro.2017.03.1032.
149. C. WANG, J. CHEN, S. QIU, W. TIAN, D. ZHANG, G.H. SU, "Performance analysis of heat pipe radiator unit for space nuclear power reactor," *Annals Nucl. Eng.*, 103, 74-84 (2017); https://doi.org/10.1016/j.anucene.2017.01.015.
150. K. K. PANDA, I. V. DULERA, A. BASAK, "Numerical simulation of high temperature sodium heat pipe for passive heat removal in nuclear reactors," *Nucl. Eng. Des.*, 323, 376-385(2017); https://doi.org/10.1016/j.nucengdes.2017.03.023.
151. PUTRA, R. S. RAMADHAN, W. N. SEPTIADI, SUTIARSO, "Visualization of the boiling phenomenon inside a heat pipe using neutron radiography," *Exp. Therm. Fluid Sci.*, 66, 13-27 (2015); https://doi.org/10.1016/j.expthermflusci.2015.02.026.
152. M. B. PARKER, "Applications of heat pipes to nuclear reactor engineering," *Iowa State University*, (1970).
153. W. A. RANKEN, "The Status of High-Temperature Heat Pipe Technology," *Los Alamos National Lab.*, (1982); https://digital.library.unt.edu/ark:/67531/metadc1190825/.
154. R. S. REID, M. A. MERRIGAN, J. T. SENA, "Review of liquid metal heat pipe work at Los Alamos," *AIP Conference Proceedings*, 217, 999 (1991); https://doi.org/10.1063/1.40058.
155. R. S. REID, M. A. MERRIGAN, "Heat Pipe Activity in the Americas – 1990 to 1995," *Los Alamos National Lab.*, (1995).
156. S. M. BRAGG-SITTON, J. E. WERNER, S. G. JOHNSON, M. G. HOUTS, D. T. PALAC, L. S. MASON, D. I. POSTON, A. L. QUALLS, "Ongoing Space Nuclear Systems Development in the United States," *Idaho National Lab.*, (2011).
157. H. JOUHARA, A. CHAUHAN, T. NANNOU, S. ALMAHMOUD, B. DELPECH, L. C. WROBEL, "Heat pipe based systems – Advances and applications," *Energy*, 128, 729-754 (2017); https://doi.org/10.1016/j.energy.2017.04.028.

158. P. MCCLURE, D. V. RAO, P. SABHARWALL, J. KENNEDY, "Database of Previous Research Supporting Microreactor Designs," *Idaho National Lab.*, (2018); https://doi.org/10.2172/1824207.
159. D. Q. WANG, B. H. YAN, J. Y. CHEN, "The opportunities and challenges of micro heat pipe cooled reactor system with high efficiency energy conversion units," *Annals Nucl. Eng.*, 149, 107808 (2020); https://doi.org/10.1016/j.anucene.2020.107808.
160. B. H. YAN, C. WANG, L. G. LI, "The technology of micro heat pipe cooled reactor: A review," *Annals Nucl. Eng.*, 135, 106948 (2020); https://doi.org/10.1016/j.anucene.2019.106948.
161. C. MUELLER, P. TSVETKOV, "A review of heat-pipe modeling and simulation approaches in nuclear systems design and analysis," *Annals Nucl. Eng.*, 160, 108393 (2021); https://doi.org/10.1016/j.anucene.2021.108393.
162. P. D. CRAVEN, M. P. SANSOUCIE, "Lightweight Damage Tolerant, High-Temperature Radiators for Nuclear Power and Propulsion," *National Aeronautics and Space Administration*, (2015); https://ntrs.nasa.gov/citations/20160008044.
163. R. W. HYERS, B. N. TOMBOULIAN, "Lightweight, High-Temperature Radiator for Space Propulsion," *National Aeronautics and Space Administration*, (2012); https://ntrs.nasa.gov/citations/20130001608.
164. W. G. ANDERSON, C. TARAU, "Variable Conductance Heat Pipes for Radioisotope Stirling Systems," *AIP Conference Proceedings*, 969, 679 (2008); https://doi.org/10.1063/1.2845031.
165. I. V. DULERA, R. K. SINHA, A. R. RAO, R. J. PATEL, "High temperature reactor technology development in India," *Prog. Nucl. Eng.*, 101, 82-99 (2017); https://doi.org/10.1016/j.pnucene.2017.04.020.
166. E. D. THROM, "Regulatory Analysis for the Resolution of Generic Issue 82, 'Beyond Design Basis Accidents in Spent Fuel Pools,'" *Nuclear Regulatory Commission,* (1989); https://www.osti.gov/servlets/purl/6113293.
167. "Long term storage of spent nuclear fuel – Survey and recommendations," IAEA-TECDOC-1293, (2002).